

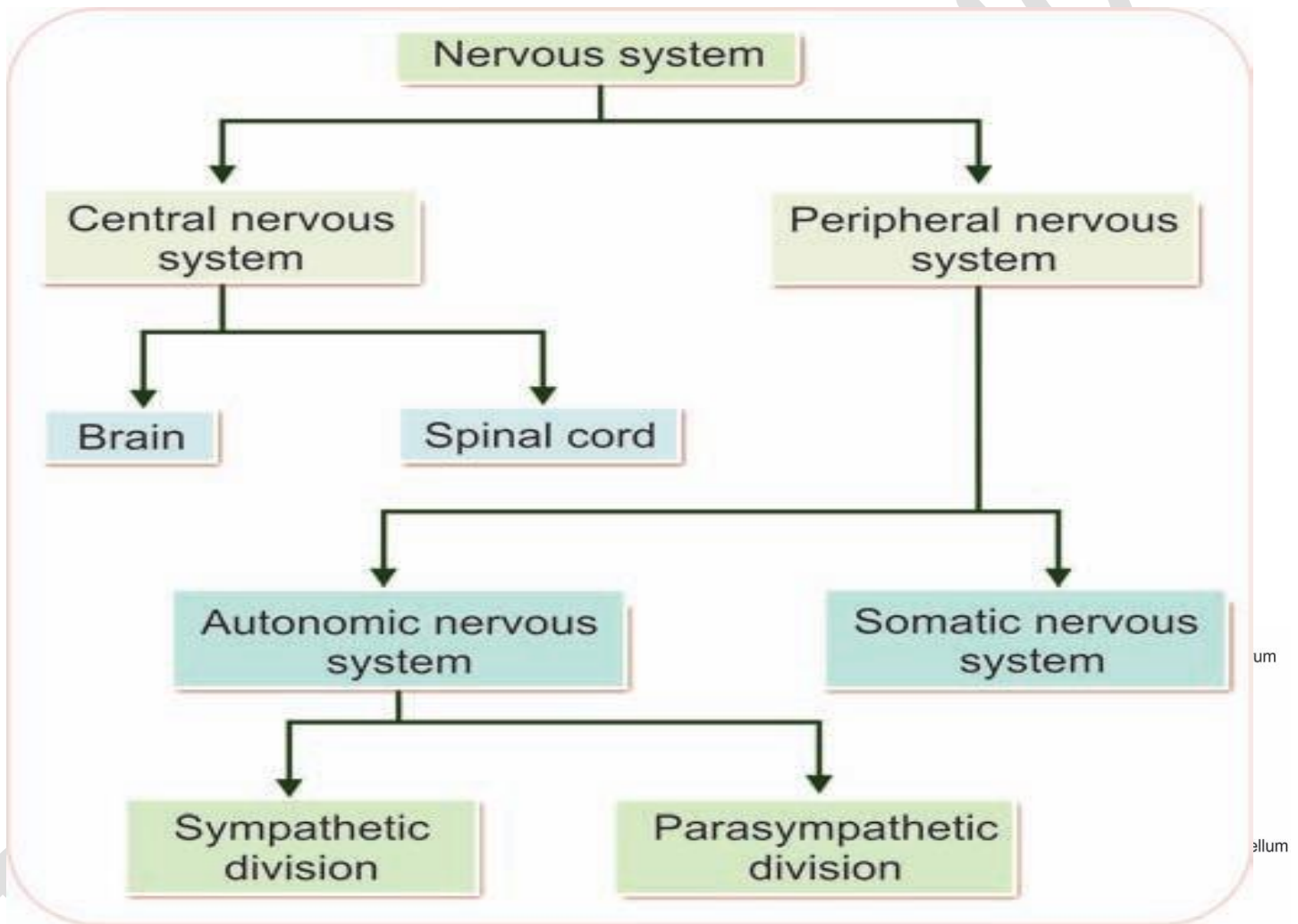
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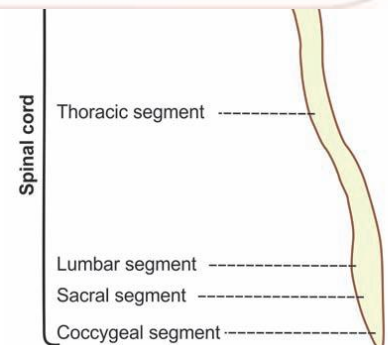
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Nervous System Divisions

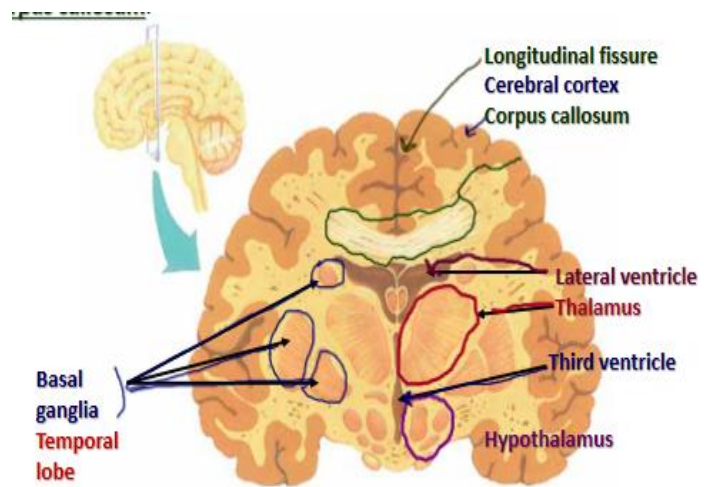


Central nervous system (CNS)—brain and spinal cord.

- ⇒ The CNS is composed of three principal structures: the brain, brainstem and spinal cord. The brain comprises two hemispheres which are joined by a band of white matter fibres known as the corpus



callosum. The grey matter known as the cerebral cortex is located on the outer surface of the hemispheres, and is composed of six layers of neurons. The cerebral cortex is divided into four anatomical regions: the frontal, temporal, parietal and occipital lobes. These each have distinct functions.



- ⇒ The CNS consists of the brain and spinal cord.
- ⇒ The functions of the CNS include body orientation and coordination, assimilation of experiences (learning), and programming of instinctual behavior.
- ⇒ The CNS contains gray and white matter.
- ⇒ The gray matter consists of either nerve cell bodies and dendrites, or of unmyelinated axons and neuroglia.
- ⇒ The white matter consists of aggregations of myelinated axons and forms nerve tracts, within the CNS.

Peripheral nervous system (PNS)—cranial nerves and spinal nerves.

- ⇒ Peripheral nervous system (PNS) is formed by neurons and their processes present in all regions of the body. It consists of cranial nerves, arising from brain and spinal nerves, arising from the spinal cord. It is again divided into two subdivisions:

- ⊕ Somatic nervous system
- ⊕ Autonomic nervous system.

Somatic Nervous System

- Somatic nervous system is concerned with somatic functions. It includes the nerves supplying the skeletal muscles. Somatic nervous system is responsible for muscular activities and movements of the body.

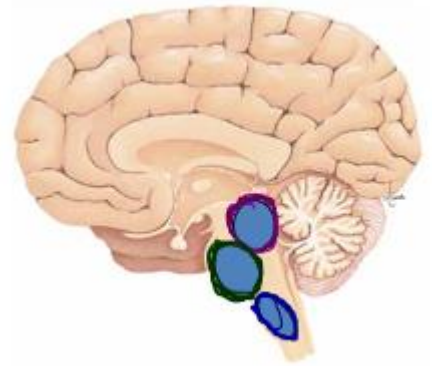
✚ Autonomic Nervous System

- Autonomic nervous system is concerned with regulation of visceral or vegetative functions. So, it is otherwise called vegetative or involuntary nervous system.
- Autonomic nervous system consists of two divisions, **sympathetic division** and **parasympathetic division**.

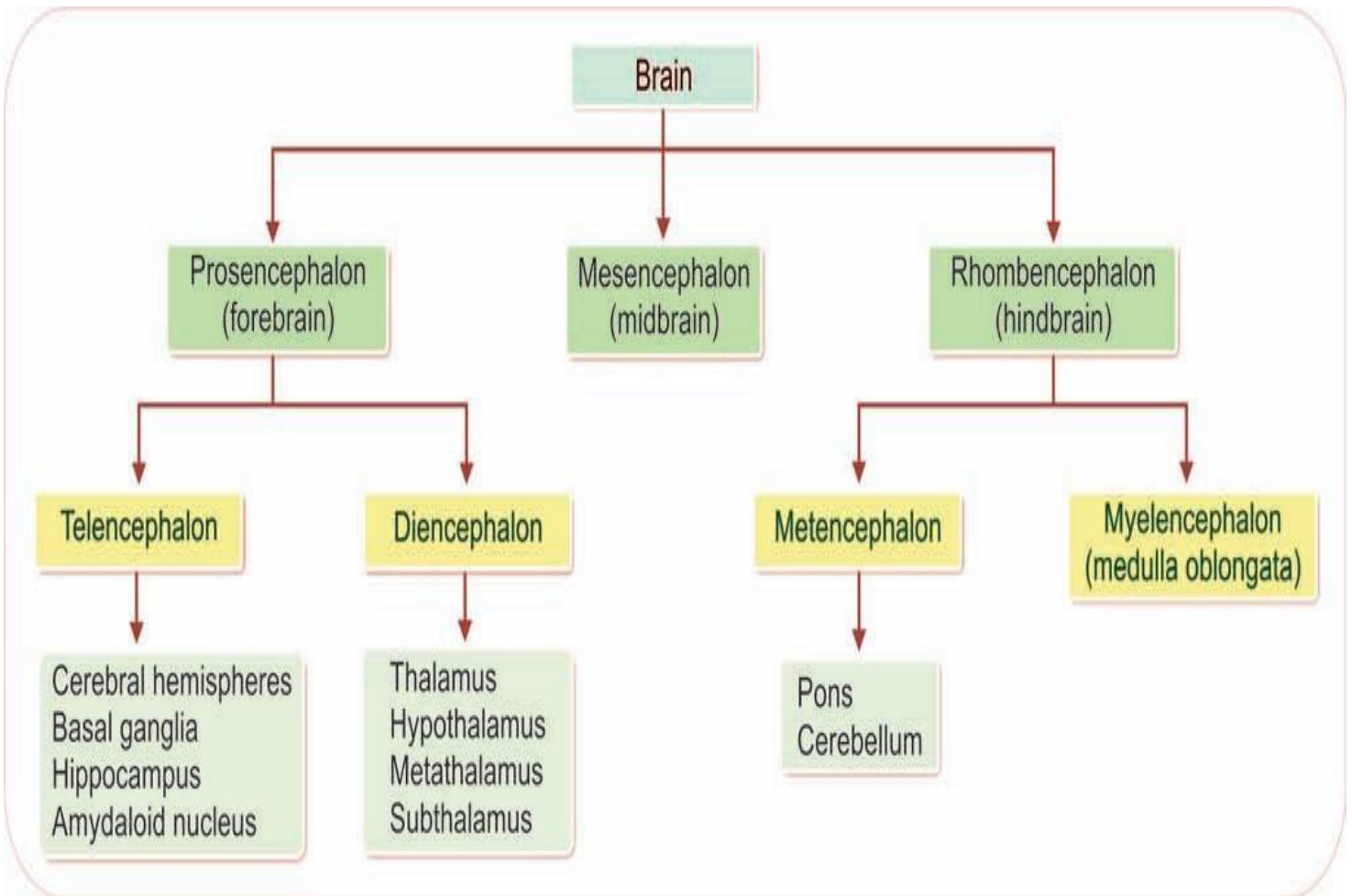
Note:

The brainstem - the elongated extension upwards within skull of the spinal cord, consisting of medulla oblongata, the pons and midbrain.

The pons and medulla = the bulb.



Embryological development of the Brain

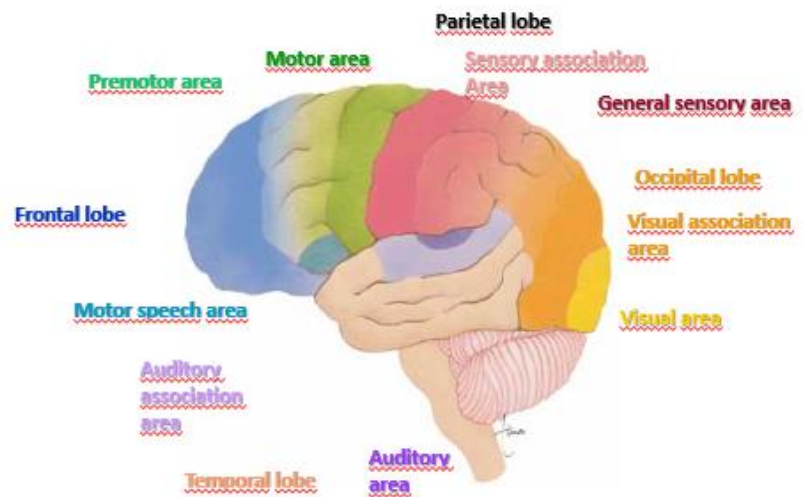


Functions of the Nervous System

1. Detect changes and feel sensations.
2. Initiate responses to changes.
3. Organize and store information.

Location and function of the lobes of the cerebral cortex.

- ⇒ The **frontal lobe** is responsible for voluntary movement, intellect, personality and memory.
- ⇒ Sensation is appreciated in the parietal cortex, which also has a major role in reading, speech and writing.
- ⇒ The **temporal lobe** has an important role in memory, mood and hearing,
- ⇒ the main function of the occipital lobe is vision.
- ⇒ The white matter beneath the cerebral cortex is composed of axons which connect the cortical neurons with neurons in other grey matter regions, including the opposite hemisphere.
- ⇒ In the Centre of the hemispheres there is a complex series of grey matter nuclei known as



1) The Basal ganglia,

- ✓ Movement co-ordination
- ✓ Tone and posture regulation

2) The Thalamus

- ✓ Movement co-ordination
- ✓ Visual co-ordination
- ✓ Sensory co-ordination
- ✓ Memory processing

3) The Hypothalamus.

- ✓ Memory processing
- ✓ Thirst, appetite and temperature control
- ✓ Autonomic regulation
- ✓ Control of pituitary endocrine function

Cerebellum

- ⇒ The cerebellum is located at the posterior surface of the brainstem; to which it is connected by white matter fibre bundles.
- ⇒ The cortex of the cerebellum lies on its outer surface, but its structure is different from the cerebral cortex.
- ⇒ Its Functions Include;
 - ✓ Movement co-ordination
 - ✓ Stance, posture and gait regulation
 - ✓ Storing and initiating learned movements

Brainstem

- ⇒ The brainstem contains many ascending and descending white matter fibre bundles which connect the spinal cord to the brain; however, it also contains many nuclei, including cranial nerves 3-12, the substantia nigra, the respiratory Centre and the vomiting Centre.

Spinal Cord

- ⇒ The spinal cord is largely composed of ascending and descending white matter fibre bundles, such as the corticospinal pathways (descending motor fibres) and the posterior columns (ascending sensory fibres).
- ⇒ The grey matter of the spinal cord is located in the centre, and contains several groups of neurones, including the anterior horn cells, which are the lower motor neurones supplying all the skeletal muscle in the trunk and limbs.
- ⇒ Motor nerve roots leave the anterior spinal cord to form peripheral motor nerves; sensory nerves from the skin, joints and organs enter the spinal cord by the posterior nerve roots, and then pass into the ascending posterior columns.
- ⇒ The skull and the vertebrae form a rigid compartment encasing the delicate brain and spinal cord. The average weight of the brain is about 1400 gm in

men and 1250 gm in women. The two main divisions of the brain— the cerebrum and the cerebellum, are quite distinct in structure. **The brain does not have lymphatic drainage.** There are 2 types of tissues in the nervous system:

- 1) **Neuroectodermal tissues** which include neurons (nerve cells) and neuroglia, and together form the predominant constituent of the CNS.
- 2) **Mesodermal tissues** are microglia, dura mater, the leptomeninges (pia-arachnoid), blood vessels and their accompanying mesenchymal cells.

⇒ Despite the structural and functional complexities of the CNS, the constituent cells can be divided into just five main groups:

1) Neurones

- The neurons are highly specialized cells of the body which are incapable of dividing after the first few weeks of birth. Thus, brain damage involving the neurons is irreversible. Neurons vary considerably in size and shape. Their size may range from the small granular cells of the cerebellum to large Betz cells of the motor cortex.
- Neurones, or nerve cells, vary considerably in size and appearance within the CNS. All possess a cell body, axons and dendrites.
- The cell body or perikaryon is easily seen by light microscopy. It contains neurofilaments, microtubules, lysosomes, mitochondria, complex stacks of rough endoplasmic reticulum, free ribosomes and a single nucleus with a prominent nucleolus.
- Some groups of neurones contain the pigment neuromelanin and are readily identifiable with the naked eye as darkly colored nuclei, e.g. in the substantia nigra.
- Neurons are the structural and functional units of the CNS, generating electrical impulses which allow rapid cell-cell communication at specialized junctions known as synapses.

- Many millions of neurons are present, arranged in layers within the cortex on the surface of the cerebellum and the cerebral hemispheres.
- Neurons are highly specialized post-mitotic cells which cannot be replaced after cell death. They are subject to unique metabolic demands, having to maintain an axon (which may be up to 1 m in length) by intracellular transport. This makes neurons particularly vulnerable to a wide range of insults, particularly hypoxia and hypoglycemia.
- Neurons are capable of generating and transmitting electrochemical impulses.
- There are many different kinds of neurons, but they all have the same basic structure.
- Some neurons are round, others oval or fusiform but the prototype of cortical neuron is pyramidal in shape.
- A neuron consists of 3 main parts: the cell body, an axon and numerous dendrites.
- **Cell Body** contains the nucleus and is essential for the continuing life of the neuron. The cell body contains a large vesicular nucleus with a single prominent nucleolus, mitochondria, and other organelles. It has abundant RER, reflecting high rates of protein synthesis. At the light microscopic level, the RER stains intensely with basic dyes and is referred to as Nissl substance. Microtubules and neurofilaments contribute to the neuronal cytoskeleton and play important roles in axonal transport. Pigment granules such as lipofuscin ("wear and tear" pigment) and melanin (found in some catecholamine-containing neurons) may be seen in the cytoplasm.
- **An axon** is a process (the term "process" here means "something that sticks out," a cellular extension) that carries impulses away from the cell body; a neuron has only one axon.

Axons are thin, cylindrical processes typically arising from the perikaryon (or from a proximal dendrite) through a short pyramidal-shaped region called the axon hillock. The cell membrane of the axon is called the axolemma, and the cytoplasm of the axon is called the axoplasm.

- **Dendrites** are processes that carry impulses toward the cell body; a neuron may have several dendrites. Dendrites are neuronal processes that receive information and transmit it to the cell body. Extensive dendritic branching serves to increase the receptive area of the neuron.

2) Glia

- Glia the special connective tissue of the central nervous system, composed of different cells, including the oligodendrocytes, astrocytes, ependymal cells (see ependyma) , and microglia , with various supportive and nutritive functions. Glial cells outnumber the neurones by between five and ten to one, and make up some 40% of the total volume of the brain and spinal cord.
- Neuroglia or glia (glia = glue) is the supporting cell of the nervous system.
- Neuroglial cells are non-excitabile and do not transmit nerve impulse (action potential). So, these cells are also called non-neural cells or glial cells.
- When compared to the number of neurons, the number of glial cells is 10 to 15 times greater. Neuroglial cells play an important role in the reaction of nerve during infection. Most commonly, neuroglial cells constitute the
- site of tumors in nervous system.
- Glial cells have a primary role in a wide range of normal functions and reactions to injury, including inflammation, repair, fluid balance, and energy metabolism.

- The size and shape of the nucleus helps in the light microscopic distinction of one glial cell type from another, as their cytoplasmic processes are often not apparent on H&E preparations and can be demonstrated only with the use of metallic impregnation, immunohistochemical, or electron microscopic methods.
- Glial cells are derived from neuroectoderm (macroglia: astrocytes, oligodendrocytes, ependymal cells)
- Glial cells are derived from bone marrow (microglia).
- The neuroglia provides supportive matrix and maintenance to the neurons.
- It includes 3 types of cells: astrocytes, oligodendrocytes and ependymal cells. Neuroglia is generally referred to as glia; the Tumours originating from it are termed gliomas, and reactive proliferation of the astrocytes being called gliosis.

☑ **Astrocytes;**

- * are physiological and biochemical support to the neurons and interactions with capillary endothelial cells to establish blood brain barrier. In case of damage to the brain, astrocytes act like fibroblasts of other tissues. The astrocytes in respond to injury undergo hyperplasia and hypertrophy termed 'gliosis' which is an equivalent of scar elsewhere in the body.
- * Astrocytes are process-bearing cells which are poorly visualised by light microscopy unless special staining techniques are used. They perform several important roles:
 - provision of a supportive framework for other cells in the CNS
 - regulation of the blood-brain barrier by astrocyte foot processes which are closely applied around cerebral capillaries.

- × Astrocytes typically have round to oval nuclei with evenly dispersed, pale chromatin; oligodendrocytes have a denser, more homogeneous chromatin in a rounder and smaller nucleus; and microglia have an elongated, irregularly shaped nucleus with clumped chromatin.
- × The cells are characterized by an eosinophilic cytoplasm and are termed gemistocytic astrocytes. Local proliferation and enhancement of the function of astrocytes leads to gliosis (the formation of a glial scar). Astrocytes are also responsible for the formation of amorphous, basophilic, rounded structures called corpora amylacea, which are aggregates of carbohydrates and proteins accumulated with normal aging. Finally, astrocytes may undergo neoplastic transformation to form astrocytomas.

☑ Oligodendrocytes

- × are so named because of their short and fewer processes when examined by light microscopy with special stains (oligo=short). The major function of oligodendrocytes is formation and maintenance of myelin. Thus, in this respect they are counterparts of Schwann cells of the peripheral nervous system. Diseases of oligodendrocytes are, therefore, disorders of myelin and myelination such as inherited leucodystrophies and acquired demyelinating diseases.
- × Oligodendrocytes are the most numerous cells in the CNS.

- × On light microscopy, they are visible as darkly staining nuclei located around neurones and nerve fibres.
- × The most important function of oligodendrocytes is the synthesis and maintenance of myelin in the CNS.
- × Oligodendroglia are glial cells that synthesize the myelin that surrounds axons. These cells have a small, dark, round nucleus and a thin rim of cytoplasm, and are situated as satellites around neurons in the gray matter and longitudinally between myelinated fibers in the white matter. They can undergo neoplastic transformation to form oligodendrogliomas and are involved in demyelinating disease.

☑ ependymal cells

- × are epithelium like and form a single layer of cells lining the ventricular system, aqueduct, central canal of the spinal cord and cover the choroid plexus. They are cuboidal to columnar cells and have ciliated luminal surface, just beneath which are present small bodies termed blepharoplasts. The ependymal cells influence the formation and composition of the cerebrospinal fluid (CSF) by processes of active secretion, diffusion, absorption and exchange. The function of cilia is not very clear but probably they play a role in the circulation of CSF. The ependymal cells respond to injury by cell loss and the space left is filled by proliferation of underlying glial fibres.
- × Ependymal cells form the single-cell lining of the ventricular system and the central canal of the spinal cord.
- × They are short columnar cells which bear cilia on the luminal surface.

- × Ependymal cells may participate in the absorption and secretion of cerebrospinal fluid (CSF).
- × Ependymal cells are glial cells that regulate the fluid transfer between the CSF and the CNS. These cells which form a single layer of cuboidal or flat cells, line the ventricular system, including the ventricular chambers, aqueduct of Sylvius, the central canal of the spinal cord, and the filum terminale. Ependymal cells can undergo malignant transformation to form ependymomas.
- × Ependymal cells, on the other hand, do have visible cytoplasm; seen with H & E, they are columnar epithelial-like cells with a ciliated/microvillous border facing the ventricular surface with pale, vesiculated nuclei located at the abluminal end of the cell.

3) microglial cells

- the nervous system counterpart of the monocyte-macrophage system. Microglial cells (or Hottel cells) are not fixed but are mobile cells. These cells are found throughout the brain and are often present close to the blood vessels. Normally, microglial cells appear as small inconspicuous cells with bean-shaped vesicular nuclei, scanty cytoplasm and long cytoplasmic processes.
- In response to injury or damage, these cells have capability to enlarge in size, proliferate and develop elongated nuclei. Microglial cells may actually assume the shape and phagocytic function of macrophages and form Gitter cells. The foci of necrosis and areas of selective hypoxic damage to the neurons are surrounded by microglial cells which perform phagocytosis of damaged and necrosed cells; this is known as neuronophagia.
- Microglia belong to the macrophage/monocyte system of phagocytic cells.

- They are normally quiescent, and inconspicuous on light microscopy, but are of major importance in reactive states, for example in inflammatory and demyelinating disorders.

4) connective tissue

- Connective tissue in the CNS is confined to two main structural groups: the meninges and perivascular fibroblasts.
- The meninges comprise the pia, arachnoid and dura mater, and the arachnoidal granulations which are the main sites of CSF absorption.
- The meninges are composed of fibroblast-like cells which also extend around meningeal and cerebral blood vessels.

5) blood vessels.

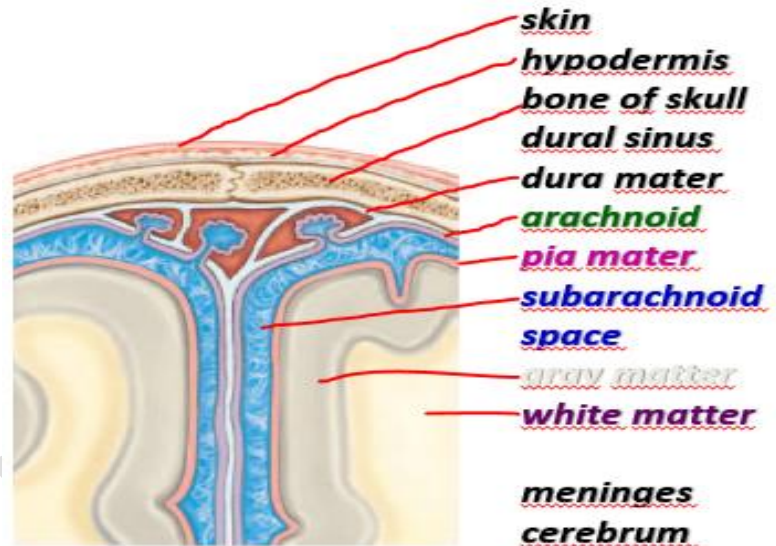
- Blood vessels in the CNS are similar in structure and function to those elsewhere in the body, with the important exception of the capillaries. The capillaries within the CNS differ from most other capillaries in several respects:
- The vessels are non-fenestrated, and tight junctions are present between adjacent cells.
- The endothelial cells possess a thick layer of cytoplasm containing numerous organelles.
- Relatively few microvilli are present on the luminal surface of the endothelial cells, and only occasional pinocytotic vesicles are present in the cytoplasm.
- The endothelial cell basement membrane is intimately surrounded by a network of astrocyte processes.
- These special structural features are important constituents of the blood-brain barrier: this is a functional unit which restricts the entry and exit of many substances-including proteins, ions, non-lipid-soluble compounds and drugs-to and from the CNS.

Meninges

- ⇒ Meninges are protective membranes that enclose the brain and spinal cord. The meninges include three layers: the dura mater, the arachnoid, and the pia mater.

DURA MATER

- ⇒ is a tough fibrous covering of the brain which is closely attached to the skull on its inner layer of endocranial periosteum.
- ⇒ In the region of spinal canal, it encloses a potential space, the epidural space, between the bone and the dura. The dura is composed of dense collagen, fused with periosteum of the skull.

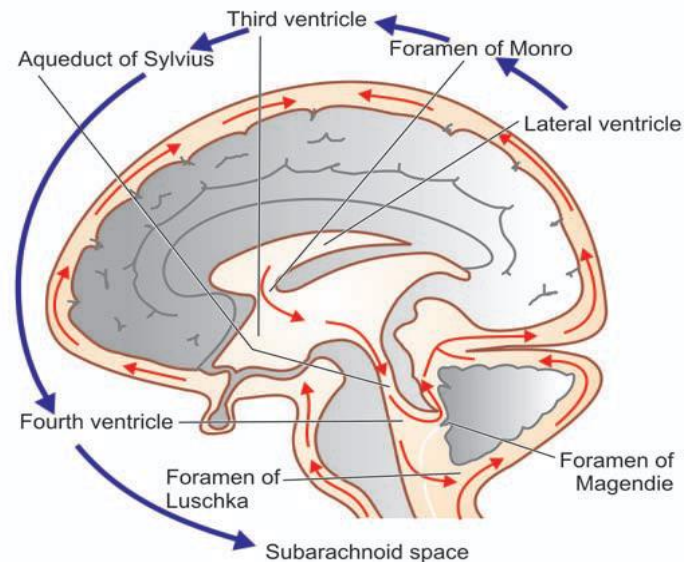


PIA-ARACHNOID (LEPTOMENINGES) (lepto=thin, slender)

- ⇒ Consisting of the pia and arachnoid mater form the delicate vascular membranous covering of the central nervous system.
- ⇒ The pia mater is closely applied to the brain and its convolutions, while the arachnoid mater lies between the pia mater and the dura mater without dipping into sulci.
- ⇒ Thus, a space is left between the two layers of leptomeninges, known as subarachnoid space, which contains the CSF.
- ⇒ The major arteries and veins run in the subarachnoid space and small nutrient arteries pass into the cortex. Extension of the subarachnoid space between the wall of blood vessels entering the brain and their pial sheaths form a circumvascular space called **Virchow-Robin space**.
- ⇒ Another important potential space is enclosed between the dura and the arachnoid membrane known as subdural space.

CSF

- ⇒ Choroid plexus is the central nervous system (CNS) organ responsible for producing cerebrospinal fluid and consists of innumerable papillae with a highly vascular core covered by cuboidal epithelium that is derived embryologically from the ependymal.
- ⇒ CSF is mainly produced by choroid plexus in the lateral, third and fourth ventricle, and a small part is formed on the surface of the brain and spinal cord. The total volume of CSF is about 120-150 ml. CS formed in the lateral ventricles flows through the **foramina of Munro** to the **third ventricle** and from there by the **aqueduct of Sylvius** to the **fourth ventricle**.
- ⇒ The fluid then passes through the **foramina of Magendie and Luschka** of the **fourth ventricle** to reach the subarachnoid space of the brain.
- ⇒ It then spreads through the subarachnoid space over the surface of the spinal cord. It is absorbed into the blood by the arachnoid villi present along the dural venous sinuses.
- ⇒ Choroid plexus cells secrete CSF and contain large quantities of mitochondria, rough endoplasmic reticulum and Golgi apparatus within the cytoplasm.
- ⇒ They form a cuboidal epithelial covering over the ventricular choroid plexus, and bear atypical microvilli.



Neurons

Neuron or nerve cell is defined as the structural and functional unit of nervous system. A nerve impulse travels along the cell membrane of a neuron, and is electrical, but where neurons meet there is a small space called a **synapse**, which an electrical impulse cannot cross. At a synapse, between the axon of one neuron and the dendrite or cell

body of the next neuron, impulse transmission depends upon chemicals called **neurotransmitters**. Neuron is similar to any other cell in the body, having nucleus and all the organelles in cytoplasm. However, it is different from other cells by two ways:

1. Neuron has branches or processes called axon and dendrites
2. Neuron does not have centrosome. So, it cannot undergo division.

CLASSIFICATION OF NEURON

Neurons are classified by three different methods.

- A. Depending upon the number of poles
- B. Depending upon the function
- C. Depending upon the length of axon.

DEPENDING UPON THE NUMBER OF POLES

Based on the number of poles from which the nerve fibers arise, neurons are divided into three types:

1. **Unipolar neurons;**

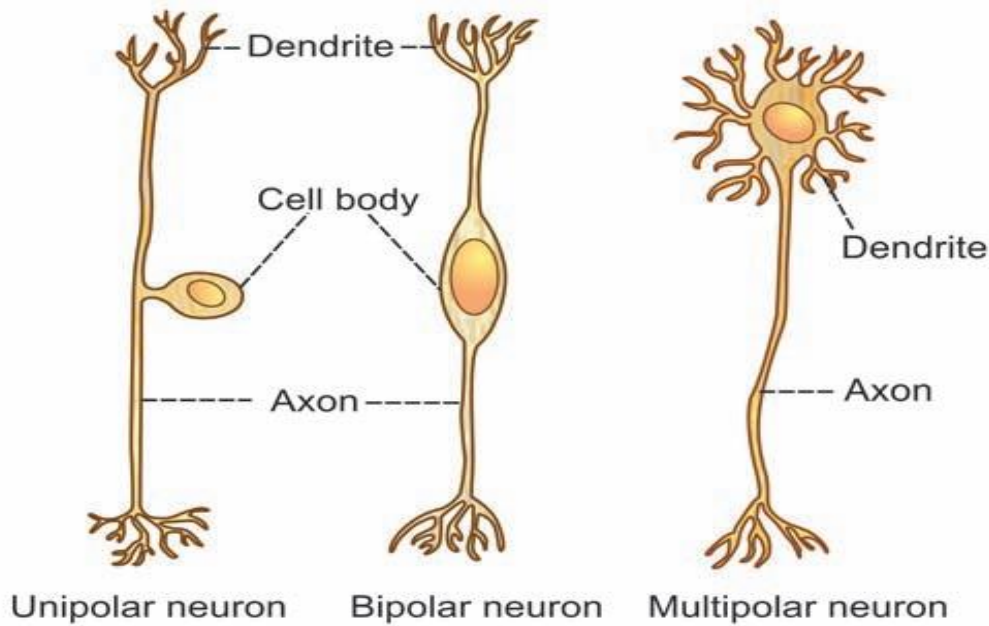
- ☑ Unipolar neurons are the neurons that have only one pole. From a single pole, both axon and dendrite arise. This type of nerve cells is present only in embryonic stage in human beings.

2. **Bipolar neurons**

- ☑ Neurons with two poles are known as bipolar neurons. Axon arises from one pole and dendrites arise from the other pole.

3. **Multipolar neurons.**

- ☑ Multipolar neurons are the neurons which have many poles. One of the poles gives rise to axon and all other poles give rise to dendrites.



DEPENDING UPON THE FUNCTION

☑ On the basis of function, nerve cells are classified into two types:

1. Motor or Efferent Neurons

Motor or efferent neurons are the neurons which carry the motor impulses from central nervous system to peripheral effector organs like muscles, glands, blood vessels, etc. Generally, each motor neuron has a long axon and short dendrites.

2. Sensory or Afferent Neurons

Sensory or afferent neurons are the neurons which carry the sensory impulses from periphery to central nervous system. Generally, each sensory neuron has a short axon and long dendrites.

3. Association neurons (Interneurons): conduct impulses from sensory to motor neurons.

A.

10102 Collection

CNS diseases (VASCULAR DISORDERS, INFLAMMATION)

The most important diseases of the central nervous system (CNS).

1. Developmental disorders
2. Physical injury of the brain and spinal cord
3. Vascular disorders
4. Infections
5. Autoimmune diseases
6. Neurodegenerative diseases
7. Nutritional, metabolic and toxic brain diseases
8. Neoplasms

1. *Discuss etiology, pathogenesis, morphology and outcomes of most common aneurysms found in the CNS.*
2. *Discuss etiology, pathogenesis, morphology and outcomes of basal ganglia and thalamus hemorrhages.*
3. *Discuss in details morphology of the main forms of brain injury caused by physical forces.*
4. *Briefly explain Duret hemorrhage.*

Haemorrhage

Haemorrhage into the brain may be **traumatic**, **non-traumatic**, or **spontaneous**. There are two main types of spontaneous intracranial haemorrhage:

☺ Intracerebral haemorrhage, which is usually of hypertensive origin.

☺ Subarachnoid haemorrhage, which is commonly aneurysmal in origin.

In addition to hypertension and rupture of an aneurysm, other causes of spontaneous intracranial haemorrhage include vascular malformations which produce mixed intracerebral and subarachnoid haemorrhage, haemorrhagic diathesis and haemorrhage into tumors.

Intracerebral (Intraparenchymal) Hemorrhage

- Spontaneous (non-traumatic) Intraparenchymal hemorrhages occur most commonly in middle to late adult life, with a peak incidence at about age 60 years.
- The most common sites of hypertensive cerebral hemorrhages.
 - i. Basal ganglia and thalamus (65%)
 - ii. Pons (15%)
 - iii. Cerebellum (10%)
- Most are caused by rupture of a small intraparenchymal vessel. Hypertension is the most common underlying cause of primary brain parenchymal hemorrhage, accounting for more than 50% of clinically significant hemorrhages. Conversely, brain hemorrhage accounts for roughly 15% of deaths among patients with chronic hypertension.
- Hypertension causes a number of abnormalities in vessel walls, including accelerated atherosclerosis in larger arteries; hyaline arteriolosclerosis in smaller vessels; and, in severe cases, proliferative changes and frank necrosis of arterioles.

Morphology.

- may originate in the putamen (50% to 60% of cases), thalamus, pons, cerebellar hemispheres (rarely), and other regions of the brain.

- When the hemorrhages occur in the basal ganglia and thalamus, they are designated **ganglionic hemorrhages** to distinguish them from those that occur in the lobes of the cerebral hemispheres, which are called **lobar hemorrhages**. Acute hemorrhages of either type are characterized by extravasation of blood with compression of the adjacent parenchyma. Old hemorrhages show an area of cavitory destruction of brain with a rim of brownish discoloration.
- **Intracerebral hemorrhage in the basal ganglia:** A hypertensive patient bled into the basal ganglia, resulting in acute severe headache, contralateral hemiparesis and rapid decline in level of consciousness. The deep cerebral nuclei (basal ganglia) and thalamus are the most common locations of intracerebral hemorrhages.
- **Intraventricular hemorrhage.** A sagittal section of the brain shows ventricular chambers filled with blood that extended from a more anterior basal ganglionic intracerebral hemorrhage. The patient died rapidly from compression of the brainstem by blood in the fourth ventricle.
- **Pontine hemorrhage.** Rupture of micro-aneurysms in the pons leads to rapid decline in level of consciousness as a result of disruption of the reticular activating system. Multiple cranial neuropathies, dysconjugate gaze, pupillary abnormalities, paralysis and dysregulation of respiration and cardiovascular systems are common. This is the second most common location of intracranial hemorrhage, and hemorrhage in the pons is often lethal.
- **Cerebellar hemorrhage.** Intracerebral hemorrhage in the cerebellum leads to acute-onset occipital headache, nausea, vomiting, vertigo and ataxia. If the hematoma expands rapidly, fatal compression of the medulla may ensue. Surgical evacuation of the cerebellar intracranial hemorrhage can be life-saving and is a neurosurgical emergency.

Ataxia (also ataxy /-si/) - the loss of full control of bodily movements.

Vertigo - a sensation of whirling and loss of balance, associated particularly with looking down from a great height, or caused by disease affecting the inner ear or the vestibular nerve; giddiness.

Vascular Disorders.

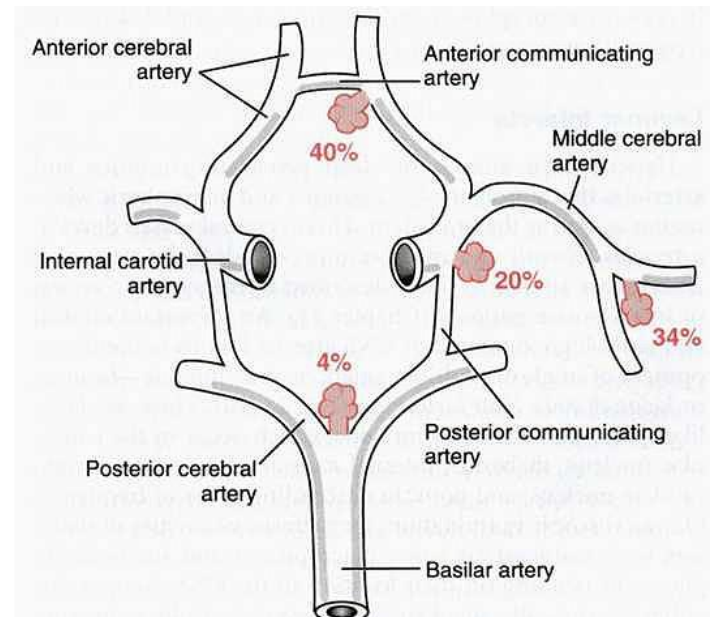
- ✓ An aneurysm is a circumscribed, segmental dilatation of an artery.
- ✓ Cerebral aneurysms may be caused by:
 - i. developmental defects of the arterial wall,
 - ii. hypertension,
 - iii. atherosclerosis,
 - iv. bacterial infection, or
 - v. trauma.
- ✓ The most important forms of intracranial aneurysms are:

Berry aneurysms:

- ✧ These small saccular aneurysms are typically found in and around the circle of Willis at the base of the brain. They develop at the site of arterial branching corresponding to the congenital weakest part of the vessels. The rupture of these aneurysms leads to a usually fatal subarachnoid and/or intraparenchymal/intraventricular haemorrhage.

Pathogenesis and Pathology:

- ✧ Berry (saccular) aneurysms are most likely caused by a developmental defect in the arterial muscle at points of bifurcation, resulting in an arterial wall composed only of endothelium, an internal elastic lamina and an adventitia.
- ✧ Intravascular pressure at this site leads to expansion and potential rupture of the aneurysm. In 20% of cases, multiple berry aneurysms occur. These aneurysms may be associated with other inherited disease, most notably adult polycystic kidney disease.



Atherosclerotic aneurysms:

- ✧ These aneurysms may involve the extraparenchymal or intracerebral arteries. Most often, they are asymptomatic and rarely rupture.
- ✧ Atherosclerotic aneurysms are situated primarily in large cerebral arteries (usually internal carotid and vertebral) and are caused by fibrous replacement of the media and destruction of the internal elastic membrane of the arterial wall. These aneurysms most commonly result in thrombosis rather than rupture.

Hypertensive micro-aneurysms:

- ✧ Long-standing hypertension leads to deposition of lipid- hyaline substances within the wall of small branches of penetrating arteries and arterioles. These deposits weaken the vessel wall and cause microscopic Charcot-Bouchard aneurysms.

Charcot-Bouchard aneurysm.

- ✧ The combination of small penetrating cerebral vessels and high perfusion pressure leads to small micro-aneurysms that may rupture, leading to intracerebral hemorrhage. Effective treatment of hypertension reduces the formation of micro-aneurysms and the frequency of intracerebral hemorrhage.

- ✧ Spontaneous cerebral hemorrhage generally occurs as a consequence of longstanding hypertension. With persistent hypertension, the walls of arterioles undergo lipid deposition and hyaline change (lipohyalinosis), followed by fibrinoid necrosis, which weakens the wall and leads to so-called Charcot-Bouchard aneurysms. Formation and rupture of these aneurysms occurs most commonly along the trunk of the vessel, rather than at points of bifurcation, and most commonly affects the following:

- Basal ganglia-thalamus (65%)**
- Pons (15%)**
- Cerebellum (10%)**

Mycotic aneurysms

- ✧ result from septic emboli that originate in infected cardiac valves and can cause arterial rupture, cerebral abscess, or meningitis.

CNS Trauma

- Trauma may result in a variety of intracranial hemorrhages, direct damage (penetrating trauma), or paralysis (spinal cord injuries).

Pathogenesis:

- A contusion describes damage to the cortex in the form of a bruise or laceration, which may be associated with the hemorrhage. Contusions occur when a rapid anteroposterior displacement of the brain occurs, such as when the skull strikes an object and is subjected to rapid, sudden deceleration. A **coup injury** (from French for blow) occurs at the site of impact, whereas a **contrecoup injury** is contralateral to the site of initial injury. The velocity of the acceleration and the abruptness of the deceleration of the head mediate the severity of a contusion.
- **Mild contusions** result in cortical bruising (local hemorrhage), whereas severe contusions may cause deep cavitory lesions within the brain that extend into the white matter and product a mass effect.

- Contusions may be life-threatening if complicated by oedema and hemorrhage, which predispose to transtentorial herniation.
- A concussion is a transient loss of consciousness due to trauma that causes a rapid torque on the brainstem, leading to a paralysis of neurons of the reticular formation. The term concussion should not be confused with cerebral contusions.

Pathology

- ◆ The foci of necrotic brain tissue and extravasated red blood cells formed by a contusion are phagocytosed by macrophages.
- ◆ Hemosiderin laden macrophages and reactive astrocytes persist indefinitely as a glial scar. In addition, diffuse axonal shearing injuries, which occur in the context of a contusion, are identified microscopically by axonal spheroids (ends of severed axons that retract) and multiple small hemorrhages.

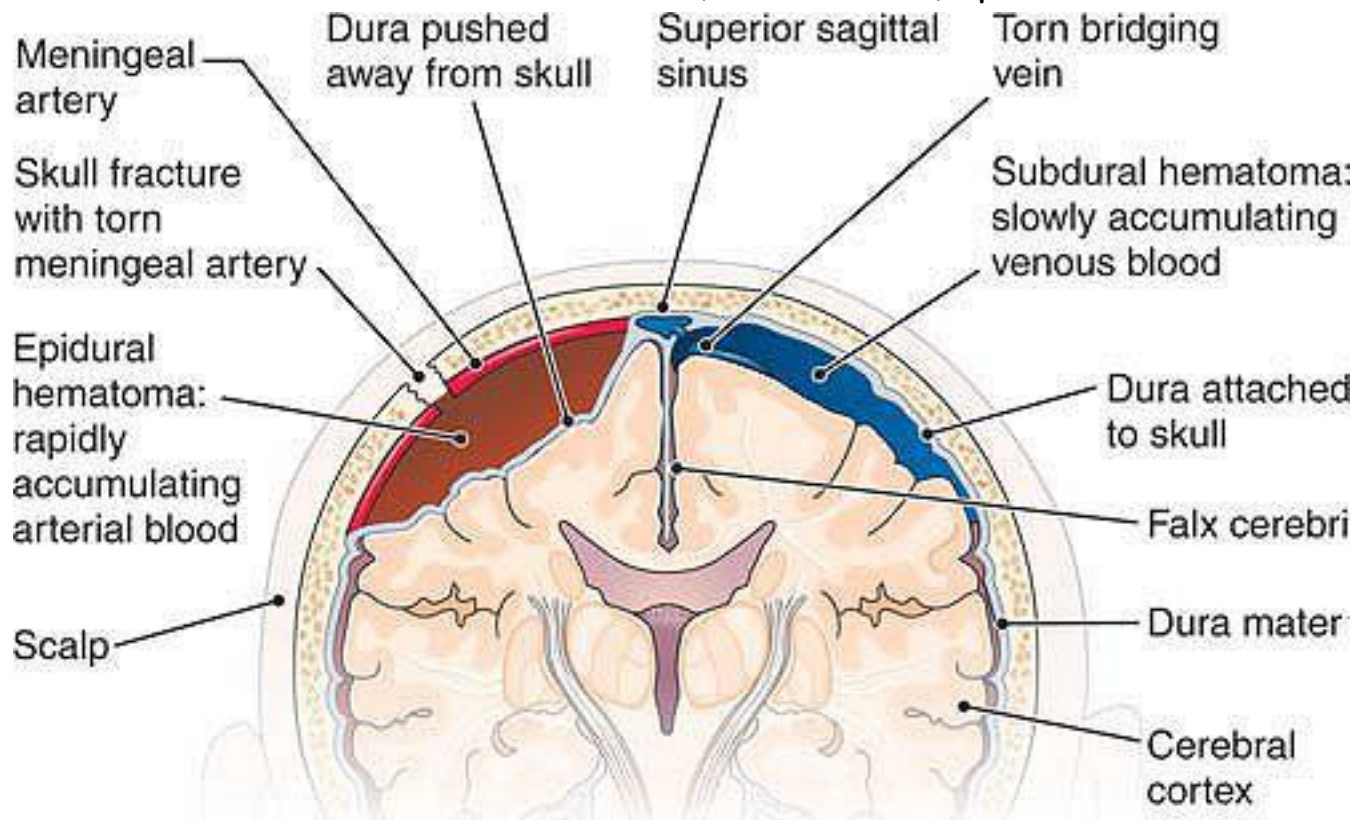
Epidural Hematoma

- ✧ An **epidural hematoma** is an arterial hemorrhage between the dura and the skull, most often resulting from skull fracture and laceration of the middle meningeal artery. Epidural hematomas are characterized clinically by a short period of consciousness (lucid interval) followed by loss of consciousness and signs of cerebral compression.

Pathogenesis

- ✧ Trauma to the temporal bone may result in transection of the middle meningeal artery, which is situated between the calvaria and dura. Damage to this artery causes a progressive accumulation of blood within the epidural space, termed an epidural hematoma.

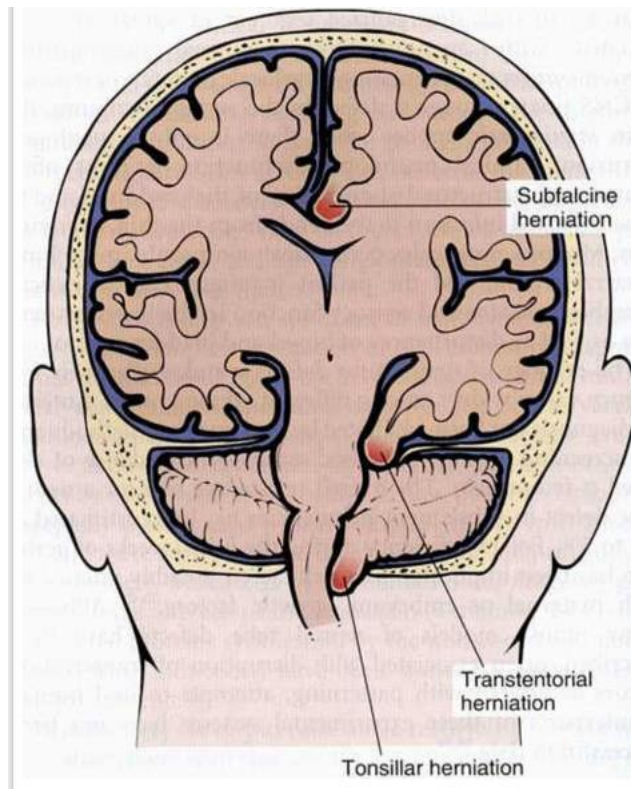
- ❖ The image below shows the difference between an epidural hematoma and subdural hematoma. Epidural and subdural hematomas. Bleeding in epidural hematoma is rapid because of skull fracture and severance of a meningeal artery. Bleeding in subdural hematoma is slow and results from tearing of veins that extend across the subdural (subarachnoid) space.



Pathology and Clinical Course:

- ❖ Within the first 4 to 8 hours, patients are often asymptomatic; however, when a critical volume of **30 to 50 mL** collects within the epidural space, patients demonstrate symptoms of a space-occupying lesion.
- ❖ As the hematoma enlarges, the large venous sinuses are compressed, leading to global cerebral hypoxia, ischemia, and confusion.

- ✧ If epidural hematomas are left untreated, they can cause transtentorial herniation, defined as displacement of the midbrain through the tentorial opening.
- ✧ This effect is evidenced by a fixed and dilated pupil on the side of the lesion and unconsciousness, as well as midbrain ischemia and necrosis. If untreated, an epidural hematoma is likely to be fatal within 24 to 48 hours.



**Major
herniations of
the brain:
subfalcine,
transtentorial,
and tonsillar.**

Subdural Hematoma

- ✧ **Subdural hematoma** refers to hemorrhage in the frontal or occipital regions of the head that causes a rapid displacement of the cerebral hemispheres against the inner aspect of the skull, such as in falls, assaults, or car accidents.
- ✧ A subdural hematoma is venous hemorrhage underneath the dura, resulting from laceration of the bridging veins. Subdural hematomas are characterized clinically by gradual signs of cerebral compression occurring hours, days, or

weeks after injury. Subarachnoid hemorrhage is commonly associated with rupture of a berry aneurysm in the circle of Willis. A transient ischemic attack is a brief episode of impaired neurologic function caused by a brief disturbance in cerebral circulation.

Pathogenesis

- ✧ Trauma can cause a shearing effect on the bridging veins within the subdural space, leading to the formation of a subdural hematoma.
- ✧ Although the subdural space can expand, most commonly blood accumulates to a volume of **25 to 50 mL**, resulting in a tamponade effect on the ruptured bridging veins. However, in some cases, venous thrombosis and ischemia may develop in the bridging veins.
- ✧ Subdural hematomas may be bilateral, owing to the nature of the injury.

Pathology and Clinical Course:

- ✧ Patients with subdural hematomas may demonstrate headaches, contralateral weakness, seizures, or lack demonstrable symptoms.
- ✧ The outcome of subdural hematomas is variable. During the first several weeks, subdural hematomas develop overlying granulation tissue, secondary to irritation between the hematoma and dura.
- ✧ Over time, the hematoma may resolve, remain static, or enlarge. Lesions that resolve contain only microscopic foci of hemosiderin-laden macrophages. Hematomas that remain static demonstrate a residual hematoma and occasionally calcification. Lesions that expand do so sporadically, often within 6 months of initial injury.
- ✧ Subarachnoid hemorrhage may occur following trauma or rupture of a berry aneurysm in the Circle of Willis, as well as in instances of vasculitis and tumours.
- ✧ It produces a sudden severe headache and photophobia, owing to meningeal irritation, which may be followed by coma.

- ✧ Often, patients experience a progressive decline in consciousness if they survive the initial hemorrhage. A subarachnoid bleed may be diagnosed by the presence of blood within the CSF when performing a lumbar puncture.

Following a bar fight, a 22-year-old man is brought unconscious to the emergency department. Several minutes earlier, he had been hit on the head with a heavy iron club and had been briefly unconscious, but had then apparently recovered. One or two minutes later, he had again lost consciousness. Which of the following is the most likely diagnosis?

- (A) Epidural hematoma
- (B) Subarachnoid hemorrhage
- (C) Subdural hematoma
- (D) Transient ischemic attack

Infections

- Inflammation is the directed tissue and its microcirculation response to noxious and injurious external and internal stimuli.
- An inflammatory response is a reaction to cell injury that can occur only in vascularized tissues of multicellular organisms.
- Inflammation is the response to injury of a tissue and its microcirculation and is characterized by the elaboration of inflammatory mediators as well as the movement of fluid and leukocytes from the blood into extravascular tissues.
- Inflammation localizes and eliminates micro-organisms, damaged cells, and foreign particles, paving the way for a return to normal structure and function.
- The CNS is prone to infection by bacteria, viruses, parasites, and prion diseases. The specific clinical findings associated with each infection are often distinct.
- Inflammation

- i. of the meninges is termed **meningitis**,
- ii. **encephalitis** is a parenchymal infection of the brain.
- iii. and inflammation of the spinal cord is named **myelitis**.

➤ Portals of entry of infection into the CNS

➤ No lymphatics enter the CNS.

1) Haematogenous spread is most common.

- × Most infectious agents reach the brain and the meninges through the arterial bloodstream. Venous blood may serve as a carrier of infectious agents from the infected periocular and perinasal tissues.

2) Trauma

- × Bacteria can be inoculated directly into the brain by bullets and sharp objects, such as a knife, or by direct entry through an open or surgical wound.

3) Local spread from paranasal sinuses, dental infections, and so forth

- × Such infection may spread from the infected middle ear or sinuses. Herpes virus residing in the trigeminal ganglion of latently infected people can also spread into the brain.

4) Via peripheral nerves

Germ	Neonates	Children	Adults
S. Pneumonia	0 – 5 %	10—20%	30 — 50%
N. Meningitis	0 – 1%	25 — 40%	10 — 50%
Strepto.	20 – 40%	2— 4%	5%
Listeria	2 — 10%	1 — 2%	5%
Gram (-) Bacilli	50 — 60%	1 — 2%	1 — 10%
Staph.	5%	1—2%	5 — 15%
Hemophilus I	0—3%	40—60%	1 — 3%

Diagnosis of meningitis (C.S.F)

	Cells	Protein	Glucose	Appearance
Bacteria. M.	Polymorphs N.	High	Low	Cloudy
Tuberculosis. M.	Lymphocyte	High	Low	Clear or opalescent
Viral . M	Lymphocyte	High	Nil	Clear
Viral Encephalitis	Lymphocyte	High	Nil	Clear

Bacterial Infections

- ☑ Follow direct spread of infection from the skull, or haematogenous spread
- ☑ Leptomeningitis is the commonest form of bacterial infection in the CNS; it occurs most frequently in children and the elderly
- ☑ Complications of bacterial meningitis include hydrocephalus, cerebral thrombophlebitis and cerebral abscess
- ☑ Cerebral abscesses are encapsulated foci of suppuration which act as space-occupying lesions
- ☑ Tuberculous infections occur as subacute meningitis or, rarely, as intracerebral tuberculomas
- ☑ CSF in bacterial meningitis contains many neutrophil polymorphs and bacteria; the fluid has high protein and low glucose concentrations.

Pathogenesis

- ⇒ Bacterial meningitis occurs when bacteria reach the meninges via bloodborne spread. Bacterial meningitis may affect the pia and arachnoid meninges (Leptomeningitis) or the dura (pachymeningitis).
- ⇒ Leptomeningitis involves infection of the CSF, which is a rich culture medium for many organisms.
- ⇒ By contrast, pachymeningitis commonly occurs when chronic sinusitis or mastoiditis extends into the external layer of the dura, often without additional spread within the CNS.
- ⇒ The most common bacteria that causes meningitis include the following:
 - i. **Escherichia coli:** This organism affects newborns, in whom a lack of transplacental maternal IgM protection against gram-negative bacteria produces a high mortality rate.
 - ii. **Haemophilus influenzae:** This gram-negative organism affects infants between the age of 3 months and 3 years.
 - iii. **Streptococcus pneumoniae:** Also known as pneumococcus, this infection occurs in adulthood and has a high incidence following basilar skull fractures.
 - iv. **Neisseria meningitidis:** This microorganism affects persons in crowded places, such as schools or barracks. It resides in the nasopharynx and early symptoms of infection include fever, malaise, and petechial rash. Severe infections may result in adrenal hemorrhages (Waterhouse-Friderichsen syndrome) or acute fulminant meningitis.
 - v. **LISTERIA MONOCYTOGENES:** Listerial meningitis is increasing in all ages and may account for up to 10% of cases of bacterial meningitis. The course is less fulminant than the other bacterial meningitides and CSF cellular responses may be lymphocyte predominant.

Pathology

- ⇒ The brain is edematous and congested.
- ⇒ The subarachnoid space is filled with a purulent exudate. The pus is most obvious in the sulci of the convexity and the subarachnoid cisterns at the base of the brain.

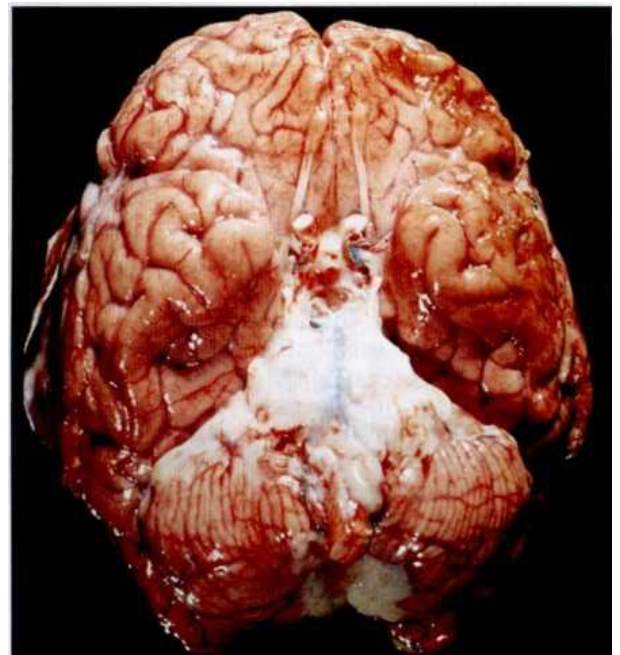
- ⇒ Microscopically the exudate in the subarachnoid space consists predominantly of live and dead neutrophils.
- ⇒ The yellow-tan exudate of acute bacterial meningitis seen here obscures the sulci.

Cryptococcal Meningitis

- ✓ Occurs Most Frequently in Immunocompromised Persons
- ✓ *Cryptococcus neoformans* causes meningitis primarily in immunocompromised hosts. Infection is often initiated by the inhalation of contaminated bird excreta.
- ✓ Lumbar puncture reveals large encapsulated spherical organisms that demonstrate a halo (capsule).
- ✓ Macroscopically, 1 mm white nodules are widely disseminated on the meninges, ependyma, and choroid plexus.
- ✓ Microscopically, organisms may be identified, although the surrounding inflammation may be minimal, with rare multinucleated giant cells and lymphocytes

Pyogenic meningitis

- ✓ Pyogenic meningitis in older infants and young children is most frequently caused by *S. pneumoniae* or *N. meningitidis*. In newborns, the most likely agents are group B streptococci, *E. coli*, and *Listeria*; in young adults, the most frequent agent is *N. meningitidis*. In older adults, especially those with impaired resistance to infection, the most common etiologic agents are *S. pneumoniae* and gram-negative rods.
- ✓ A thick layer of suppurative exudate covers the brain stem and cerebellum and thickens the leptomeninges. (image)



- ✓ Purulent exudate in the subarachnoid space is characteristic. CSF findings of diagnostic significance include:
 - a) Numerous neutrophils
 - b) Decreased glucose (less than two-thirds of the serum glucose concentration)
 - c) Increased protein

Etiology

- ✓ In neonates and infants, pyogenic meningitis is most frequently caused by group B streptococci, Escherichia coli, and Listeria.
- ✓ In older infants, children, and young adults, the disease is most frequently caused by Streptococcus pneumoniae (pneumococcus) and Neisseria meningitidis.
- ✓ N. meningitidis occurrence may be sporadic or epidemic and may be accompanied by meningococcemia secondary to a primary infection in the nasopharynx.
- ✓ Meningococcemia can also be associated with purpuric skin lesions and is sometimes complicated by the **Waterhouse-Friderichsen syndrome** (haemorrhagic destruction of the adrenal cortex, acute hypocorticism with circulatory collapse, and disseminated intravascular coagulation).

Acute Meningitis

- ✓ Acute meningitis caused by bacteria resistant to treatment may become chronic, but even in this circumstance, the exudate remains purulent.
- ✓ This prolonged purulent meningitis must be distinguished clinically and for treatment purposes from chronic meningitis caused by pathogens that typically cause chronic diseases.
- ✓ Such chronic meningitis typically has an insidious onset and may last weeks or months.

- ✓ The brain stem shows linear midline hemorrhages, called Duret hemorrhages. The acute bacterial meningitis led to brain swelling with edema and subsequent herniation with Duret hemorrhages in the pons.

Chronic Meningitis

- ✓ Chronic meningitis may be a feature of:
 - Tuberculosis (prevalent in developing countries)
 - Fungal diseases, especially in immunosuppressed people suffering from acquired immune deficiency syndrome (AIDS)
 - Lyme disease caused by *Borrelia burgdorferi*
 - Syphilis caused by *Treponema pallidum* (CNS infection is a feature of tertiary syphilis.)
- ✓ The pathologic findings in various forms of chronic meningitis vary depending on the causative pathogen.
- ✓ The infection is usually more circumscribed than in acute meningitis and may be localized to parts of the brain.
- ✓ For example, tuberculous meningitis typically involves the basal cisterns and the lateral sulci that contain gelatinous whitish gray material. Histologically, these lesions contain caseating granulomas.

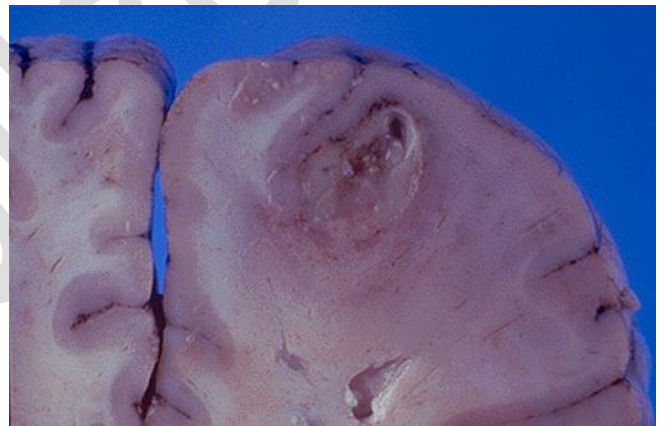
The typical clinical features of acute and chronic meningitis

- Headache, usually of sudden onset and quite severe; often localized to the forehead
- Fever (high, over 40°C in acute bacterial, but less prominent in aseptic meningitis)
- Prodrome often seen in viral diseases, but bacterial infection may have a sudden onset
- Nausea and vomiting
- Stiff neck
- Mental confusion, somnolence, photophobia, seizures, and neurologic symptoms
- Kernig sign (hip flexion induces knee pain)

- Brudzinski sign (upon flexion of the neck, there is spontaneous similar movement of the hips and knees)
- Increased cerebrospinal fluid (CSF) pressure with changes in the biochemical composition of CSF.
- Hydrocephalus resulting from obstruction of CSF due to scarring of meninges
- Neurologic defects due to destruction of underlying brain
- Epilepsy due to focal damage of the brain or scarring of meninges
- Abscess formation in the brain or subdural spaces (especially in children)

Cerebral Abscess

- Abscesses usually result from hematogenous spread of bacterial infection, but may also occur from direct penetrating trauma or extension from adjacent infection in sinuses.
- Its transmitted (***The pathogenesis and pathology of cerebral abscesses.***) as follows;
 - ✓ Localized suppuration of the brain is caused by pyogenic bacteria (*Streptococcus pneumoniae* or *Staphylococcus aureus*) presenting clinically as a destructive, spaceoccupying lesion.
 - ✓ It may be solitary (developing from local foci of infection) or multiple (in sepsis and septic embolism from endocarditis). It is also a complication of suppurative meningitis.
 - ✓ Solitary abscesses may evolve from suppurative infection of nasal sinuses (frontal lobe), the middle ear, and mastoid bone (temporal lobe or cerebellum).
 - ✓ Solitary hematogenous abscess is usually a consequence of septic embolization from infected endocardial valves. Emboli cause a cerebral infarct, which then becomes infected and transforms into an abscess.



Tuberculoma

- Tuberculosis occurs as tuberculosis of the brain substance or as tuberculous meningitis. It is secondary to tuberculous infection occurring elsewhere in the body.
- It is a focus of caseous necrosis present in the pons and midbrain (arrow). A photomicrograph shows caseous necrosis, macrophages and Langhans giant cells in a tuberculoma. If the tuberculoma ruptures into the cerebrospinal fluid, tuberculous meningitis will ensue.

Viral Infections

Introduction

- Infections spread to the CNS by the haematogenous route or by retrograde neural transport
- Viral meningitis is a common, self-limiting illness with characteristic CSF changes
- Encephalitis is less frequent, but may result in death or severe disability
- Reactivation of a latent viral infection (e.g. herpes zoster) may damage the CNS
- CNS involvement in HIV infection is common and often accompanied by other viral, bacterial or parasitic infections
- 'Slow' virus infections are responsible for subacute spongiform encephalopathy, a rare cause of dementia
- Acute disseminated encephalomyelitis, a demyelinating disorder, may result from a virus-induced immune reaction

Viral meningitis (also known as aseptic meningitis)

- may occur as part of a systemic viral disease (e.g., varicella, mumps, and measles),
- or it may be limited to the CNS. The latter may be sporadic (e.g., herpes virus infection) or epidemic (e.g., arboviruses transmitted by arthropods such as eastern or western equine encephalitis),

- and is often associated with encephalitis (i.e, meningoencephalitis).
- Infection of the meninges may be the most common viral disease of the CNS, but, unlike bacterial meningitis, it is usually benign and resolves without sequelae.

Viral Encephalitis It is often associated with meningitis (meningoencephalitis).

- Most infections are hematogenous.
- It may be diffuse or preferentially localized to specific areas (e.g., poliomyelitis affects the anterior horns of the spinal cord, and herpes simplex virus affects the temporal lobe).
- Some infections occur in previously healthy people (e.g., epidemic arthropod-borne encephalitis, rabies, and poliomyelitis), but many are found more often or exclusively in immunosuppressed people (e.g., CMV encephalitis).
- Virus may remain in latent form (e.g., herpes simplex virus in trigeminal nerve) and become reactivated by events that change the interaction between the host and the virus.
- **Ka Pathology kenda so;**
 - ✓ Perivascular infiltrates of lymphocytes (perivascular cuffing)
 - ✓ Neural injury followed by phagocytosis of damaged and killed cells by microglia cells and macrophages (neuronophagia and microglial nodules)
 - ✓ Intraneural inclusions are seen in some diseases but not in others
 - ✓ Edema and widening of the perivascular spaces
 - ✓ Concomitant meningitis (meningoencephalitis)
- **Then tuma CLINICAL FEATURES tule enda so:**
 - ✓ The onset of most viral encephalitides is usually abrupt.
 - ✓ The more specific neurologic deficits (e.g., paralysis of poliomyelitis or difficulty in swallowing in rabies) reflect the localization of the infection. Although most encephalitides run a rapid course, the tempo can vary. For example, the clinical course of subacute sclerosing panencephalitis may last

years. Herpes simplex and varicella-zoster viruses may reside latently in sensory ganglia for years, only to be reactivated decades after initial infection.

Rabies virus

- caused by an enveloped, single-stranded RNA virus (rhabdovirus).
- The virus is transmitted by the bite (infected saliva) of carnivore wild animals (e.g., foxes, bats, and dogs).
- The virus reaches the CNS by centripetal axoplasmic flow, and it reaches the viscera (salivary glands) by intraaxonal centrifugal propagation.
- Lymphocytic perivascularitis is seen around small vessels within the brain stem, basal ganglia, or hypothalamus.
- Negri bodies are found most prominently in the hippocampus, Purkinje cells, and the brain stem.
- Infection is accompanied by almost no glial inflammatory reaction in the brain.
- Clinically, there is generalized encephalopathy, presenting with irritability, seizures, contracture of the pharyngeal muscles (hydrophobia—inability to swallow water), and, terminally, delirium.
- Preventive immunization and serial vaccination of infected people are effective. After the virus reaches the brain, death is inevitable.

Herpes Simplex Virus

- Most Common Cause of Nonepidemic Encephalitis
- Herpes simplex virus type 1 (HSV-1) causes cold sores on the lips and can retrogradely infect the gasserian ganglion via the mandibular nerve trunk.
- Infection of the CNS primarily affects the temporal lobes and results in swollen, hemorrhagic, necrotic brain parenchyma, with perivascular lymphocytic cuffing.
- Infected neurons and glial cells contain small, eosinophilic, nuclear inclusions, which can be specifically identified by immunostains for HSV-1.

- Herpes simplex virus type 2 (HSV-2) is a sexually transmitted disease that causes vesicular lesions of the vagina and penis (genital herpes).
- Transmission of HSV-2 to newborns during passage through the birth canal results in severe neonatal encephalitis, with liquefactive necrosis of the cerebrum and cerebellum.
- Grossly, there are swollen hemorrhagic necrotic temporal lobes. The patient may experience memory disturbances and complex partial seizures as a result of this selective involvement of the temporal lobes.
- Microscopically, the specimen exhibits pronounced perivascular lymphocytic inflammation. This finding indicates that active inflammation is present, but is not etiologically specific.
- The infected neurons display intranuclear, eosinophilic viral inclusions (Cowdry A inclusions) that fill the nuclei (arrows). The presence of these findings is extremely valuable in guiding diagnostic evaluation as a limited number of viruses produce Cowdry A inclusions.

Cytomegalo Virus

- CMV infection is one of the congenital TORCH infections, and it can become widely disseminated to affect the CNS. Periventricular necrosis is characteristic of CMV infection. Failure of the heart in utero causes hydrops fetalis. Listeriosis can produce focal micro-abscesses in a variety of organs, but there is generally minimal necrosis. HIV infections have no significant CNS findings in the perinatal period. Herpes simplex infection of neonates typically occurs during passage through the infected birth canal of the mother, not in utero.
- TORCH complex is a group of infections transmitted from the mother to the fetus with similar clinical manifestations.
 - i. TORCH stands for TOxoplasma, Rubella, Cytomegalovirus, and Herpes simplex virus (and others, such as congenital syphilis).
 - ii. TORCH infections can lead to a syndrome characterized by microcephaly, chorioretinitis, central nervous system (CNS) calcifications, petechial rash, hepatosplenomegaly, and thrombocytopenia.

Poliomyelitis.

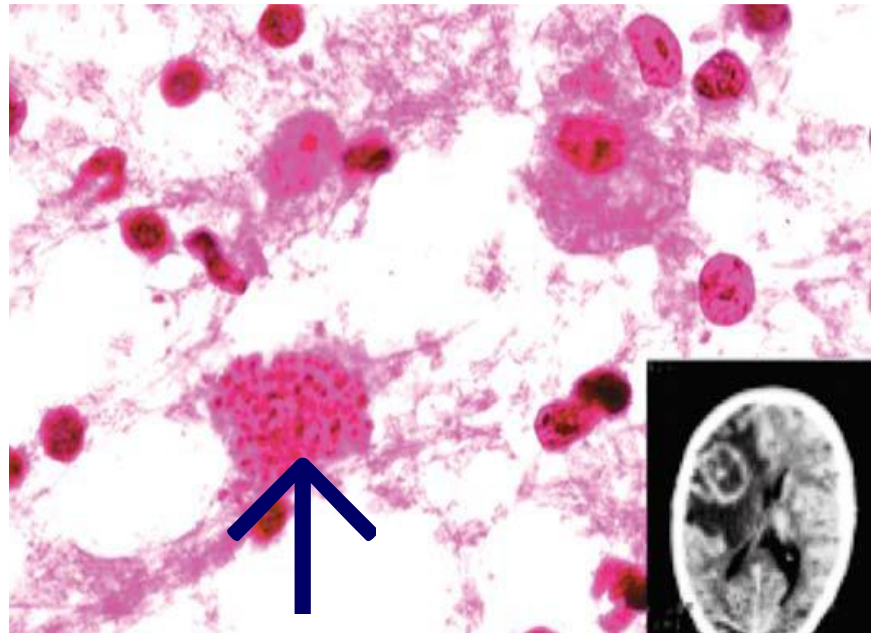
- It is extremely rare in the United States; the disease has been fully controlled by preventive immunization at an early age.
- Enterovirus infection by the fecal-oral route begins as a gastroenteritis.
- The virus spreads hematogenously to the spinal cord, causing meningoencephalitis.
- The motor neurons of the anterior horn are preferentially infected by virus and destroyed after a short phase of aseptic meningoencephalitis.
- Severe flaccid paralysis of variable severity ensues.
- Unless there is severe bulbar paralysis, most patients will have only limited consequences (motor defects).
- Severe cases are lethal. Death may be caused by involvement of respiratory muscles.

Parasitic Central Nervous System Infections

Introduction

- Toxoplasmosis is a ubiquitous protozoan to which most of us have protective immunity. Immunocompromised patients lose the ability to limit proliferation of these organisms.
- Small comma like tachyzoites and large polyorganismal cysts (bradyzoites) are seen in association with chronic inflammation, tissue necrosis and vasculitis. Toxoplasmosis is the most frequent cause of multiple intracranial masses in AIDS patients.
- Toxoplasma infection. Abscesses are present in the putamen and thalamus.

- Toxoplasmosis in a human immunodeficiency virus (HIV) patient. This previously asymptomatic patient presented with an irregularly enhancing mass with surrounding edema that was initially thought to be a high-grade neoplasm. *Toxoplasma gondii* bradyzoites (arrow) are present in a necrotic inflammatory background. Toxoplasmosis is the most common mass lesion in patients with AIDS and is an indicator disease of HIV infection.



CNS malaria

- most commonly caused by *Plasmodium falciparum*.
- During attacks of cerebral malaria, the CSF shows elevated protein and pressure but pleocytosis is uncommon.
- In fatal cases, the brain is diffusely swollen and may be otherwise unremarkable, but one may see microinfarcts with gliosis (Durck granulomata) in white matter or numerous small hemorrhages.
- Both the infarcts and microhemorrhages may be caused by obstruction of blood flow in small vessels as a result of parasitemia. Severity of cerebral malaria is also related to tumor necrosis factor release from the host immune system.

Tumors of the CNS

Introduction

- The majority of neoplasms within the CNS are actually metastatic lesions, although a variety of primary neoplasms can occur.

- Tumors of the nervous system have several unique characteristics that set them apart from neoplastic processes elsewhere in the body.
- **First**, the distinction between benign and malignant lesions is less evident in the CNS than in other organs. Some glial tumors with histologic features of a benign neoplasm, including low mitotic rate, cellular uniformity, and slow growth, may infiltrate large regions of the brain, thereby leading to serious clinical deficits and poor prognosis.
- **Second**, the ability to surgically resect infiltrating glial neoplasms without compromising neurologic function is limited.
- **Third**, the anatomic site of the neoplasm can have lethal consequences irrespective of histologic classification; for example, a benign meningioma, by compressing the medulla, can cause cardiorespiratory arrest.
- **Finally**, the pattern of spread of primary CNS neoplasms differs from that of other tumors: even the most highly malignant gliomas rarely metastasize outside the CNS. The subarachnoid space provides a pathway for spread, so seeding along the brain and spinal cord can occur in highly anaplastic as well as in well-differentiated neoplasms that extend into the CSF pathways.

General characteristic.

- ☺ Most tumors are intracranial; tumors of the spinal cord are much less frequent.
- ☺ In adults, the majority of intracranial tumors are supratentorial.
- ☺ In children, the majority of intracranial tumors are infratentorial.
- ☺ CNS tumors are the second most common form of malignancy in children (only leukemia is more frequent).
- ☺ Primary malignant CNS tumors rarely metastasize.
- ☺ Benign intracranial tumors can result in devastating clinical consequences due to compression phenomena.
- ☺ Metastatic tumors to the brain are found more frequently than primary intracranial neoplasms; in order of frequency, the most common primary intracranial tumors in adults are glioblastoma multiforme, meningioma, and

acoustic neuroma. The most common primary intracranial tumors in children are cerebellar astrocytoma and medulloblastoma.

HISTOGENETIC CLASSIFICATION OF TUMOR CELLS

↓ Cell of Origin	↓ Tumor
Astrocyte	Astrocytoma Glioblastoma multiforme
Oligodendroglia cell	Oligodendroglioma
Ependymal cell	Ependymoma
Choroid plexus cell	Choroid papilloma
Neuronal precursor cell	Medulloblastoma
Meningeal arachnoid cell	Meningioma
Vascular cell	Hemangioblastoma
Lymphoid cell	Lymphoma
Schwann cells	Schwannoma
Embryonic remnants	Craniopharyngioma Germ cell tumor

Another form of classification may include;

TUMOURS OF MENINGES	POORLY- DIFFERENTIATED	TUMOURS OF NEURONS AND NEUROGLIA	TUMOURS OF NEURONS
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AND EMBRYONAL TUMOURS

⊗ Meningioma	⊗ Medulloblastoma	⊗ Ganglioglioma	⊗ Neuroblastoma
⊗ Meningeal sarcoma	⊗ Neuroblastoma		⊗ Ganglioneuroblastoma
			⊗ Ganglioneuroma

TUMOURS OF NEUROGLIA (GLIOMAS)

⊗ Astrocytoma
⊗ Oligodendroglioma
⊗ Ependymoma
⊗ Choroid plexus papilloma

NERVE SHEATH TUMOURS

⊗ Schwannoma (neurilemmoma)
⊗ Neurofibroma
⊗ Malignant nerve sheath tumour

OTHER PRIMARY INTRAPARENCHYMAL TUMOURS

⊗ Haemangioblastoma
⊗ Primary CNS lymphoma
⊗ Germ cell tumours

MISCELLANEOUS TUMOURS

⊗ Malignant melanoma
⊗ Craniopharyngioma
⊗ Pineal cell tumours
⊗ Pituitary tumours

TUMOUR-LIKE LESIONS

⊗ epidermal cyst,
⊗ dermoid cyst
⊗ colloid cyst

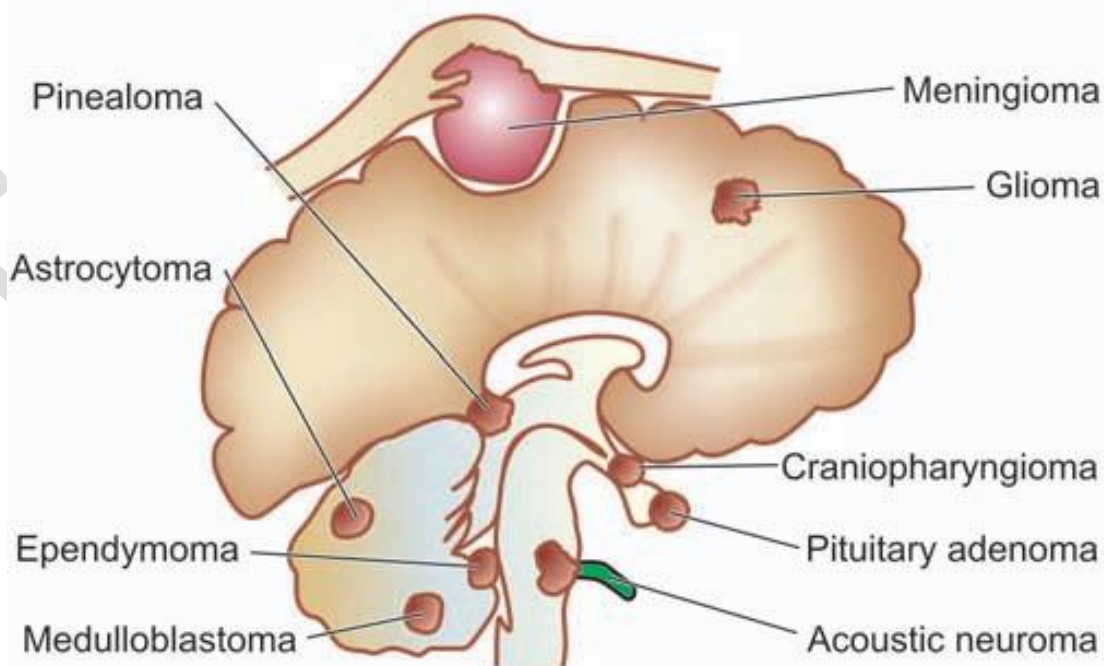
Classes of Brain tumors

- The four major classes of brain tumors are:
 - a. Gliomas
 - b. Neuronal tumors
 - c. Poorly differentiated neoplasms

d. Meningiomas

Gliomas

- ✓ Gliomas, derived from glial cells, include astrocytomas, oligodendrogliomas, and ependymomas.
- ✓ The term glioma is used for all tumours arising from neuroglia, or more precisely, from neuroectodermal epithelial tissue. Gliomas are the most common of the primary CNS tumours and collectively account for 40% of all intracranial tumours. They include tumours arising:
 - ✧ from astrocytes (astrocytomas and glioblastoma multiforme);
 - ✧ from oligodendrocytes (oligodendroglioma);
 - ✧ from ependyma (ependymoma); and
 - ✧ from choroid plexus (choroid plexus papilloma).
- ✓ Gliomas may be well-differentiated or poorly-differentiated. However, gliomas are never truly well-demarcated or encapsulated and thus all grades of gliomas infiltrate the adjacent brain tissue. Gliomas are disseminated to



other parts of the CNS by CSF but they rarely ever metastasise beyond the CNS.

Astrocytoma

- ✧ Several different categories of tumors derived from astrocytes are recognized, including fibrillary astrocytoma, glioblastoma, pilocytic astrocytoma, and pleomorphic xanthoastrocytoma as well as rarer types.
- ✧ Most gliomas are astrocytomas and as in most gliomas, it is difficult to tell where the margin is.
- ✧ Astrocytomas are the most common primary brain tumors. They can be divided based on their infiltration into the surrounding brain parenchyma.
- ✧ Astrocytomas that do not infiltrate the brain include **pilocytic astrocytomas**, **pleomorphic xanthoastrocytomas**, and **subependymal giant cell astrocytomas**.
- ✧ Diffuse astrocytomas can be further subdivided based on grade.
 - a) Low-grade fibrillary astrocytomas
 - b) Anaplastic astrocytomas
 - c) Glioblastoma multiforme (GBM)
- ✧ This is the most common primary intracranial malignancy.
- ✧ Peak occurrence is in the late middle-age group.
- ✧ This neoplasm is associated with marked anaplasia and pleomorphism; pronounced vascular changes with endothelial hyperplasia occur. Areas of necrosis and hemorrhage are surrounded by a "pseudopalisade" arrangement of tumor cells.

Fibrillary (Diffuse) Astrocytomas

- ❖ These, together with glioblastomas, account for about 80% of adult primary brain tumors. Usually found in the cerebral hemispheres, they may also occur in the cerebellum, brainstem, or spinal cord, most often in the fourth through sixth decades.
- ❖ The most common presenting signs and symptoms are seizures, headaches, and focal neurologic deficits related to the anatomic site of involvement. Fibrillary astrocytomas show a spectrum of histologic differentiation that correlates well with clinical course and outcome.

Morphology.

- ❖ Among the diffuse fibrillary astrocytomas, tumors can be well differentiated (astrocytoma) or less differentiated (higher-grade), ranging from anaplastic astrocytoma to glioblastoma.
- ❖ The macroscopic appearance of diffuse fibrillary astrocytoma is that of a poorly defined, gray, infiltrative tumor that expands and distorts the invaded brain.

Glioblastoma

- ❖ Glioblastoma (previously called glioblastoma multiforme) has a histologic appearance similar to anaplastic astrocytoma with the additional features of necrosis and vascular or endothelial cell proliferation, each of which often has a stereotypic appearance.
- ❖ Necrosis in glioblastoma, often in a serpentine pattern, occurs in areas of hypercellularity with highly malignant tumor cells crowded along the edges of the necrotic regions, producing a histologic pattern referred to as pseudopalisading.
- ❖ Glioblastoma multiforme may appear as a necrotic, hemorrhagic, infiltrating mass. It is also a Foci of necrosis with pseudopalisading of malignant nuclei.
- ❖ GBM is the most common primary intracranial neoplasm. This brain tumor has a very poor prognosis, with death occurring in less than 1 year.

- ❖ Glioblastoma multiforme, the most frequently occurring primary neoplasm of the CNS. Characteristic features include the intracerebral location, prominent hemorrhage and necrosis, and the pseudopalisading appearance of the pleomorphic tumor cells.

Pilocytic Astrocytoma

- ❖ Pilocytic astrocytomas are distinguished from the other types by their pathologic appearance and relatively benign behavior.
- ❖ They typically occur in children and young adults and are usually located in the cerebellum but may also appear in the floor and walls of the third ventricle, the optic nerves, and occasionally the cerebral hemispheres.
- ❖ Pilocytic astrocytoma may appear in the cerebellum with a nodule of tumor in a cyst.

Oligodendroglioma

- ✧ Oligodendroglia are glial cells that synthesize the myelin that surrounds axons. These cells have a small, dark, round nucleus and a thin rim of cytoplasm, and are situated as satellites around neurons in the gray matter and longitudinally between myelinated fibers in the white matter. They can undergo neoplastic transformation to form oligodendrogliomas and are involved in demyelinating disease.
- ✧ These tumors constitute 5% to 15% of gliomas and are most common in the fourth and fifth decades.
- ✧ Patients may have had several years of neurologic complaints, often including seizures.
- ✧ The lesions are found mostly in the cerebral hemispheres, with a predilection for white matter.
- ✧ Oligodendrocytes are the most numerous cells in the CNS.
- ✧ On light microscopy, they are visible as darkly staining nuclei located around neurones and nerve fibres.

- ✧ The most important function of oligodendrocytes is the synthesis and maintenance of myelin in the CNS.
- ✧ Oligodendroglioma presents as a slow-growing tumor in the middle-age group and typically arises in the cerebral hemispheres

Morphologic characteristics include:

- i. Closely packed cells with large round nuclei surrounded by a clear halo of cytoplasm ("fried egg" appearance)
- ii. Tumor divided into groups of cells by delicate capillary strands
- iii. Foci of calcification

Clinical Features.

- ✧ In general, patients with oligodendrogliomas have a better prognosis than do patients with astrocytomas.
- ✧ Current treatment with surgery, chemotherapy, and radiotherapy has yielded an average survival of 5 to 10 years.
- ✧ Patients with anaplastic oligodendroglioma have a worse prognosis, although there are subgroups of tumors, as defined by genetic alterations, that respond well to therapy.

Ependymoma and Related Paraventricular Mass Lesions

- ✧ Ependymomas most often arise next to the ependyma-lined ventricular system, including the oft-obliterated central canal of the spinal cord.
- ✧ Most frequently occurs in the fourth ventricle.
- ✧ Peak incidence is in childhood and adolescence.
- ✧ Histologic characteristics include tubules or rosettes with cells encircling vessels or pointing toward a central lumen; tumor cells characteristically demonstrate blepharoplasts, rod-shaped structures near the nucleus representing basal bodies of cilia.

- ✧ Results may include papillary growths that obstruct flow of CSF and lead to hydrocephalus.
- ✧ Ependymomas are benign histologically.
- ✧ In the first two decades of life, they typically occur near the fourth ventricle and constitute 5% to 10% of the primary brain tumors in this age group.

Morphology.

- ✧ In the fourth ventricle, ependymomas are typically solid or papillary masses extending from the floor of the ventricle.
- ✧ Microscopically, the hallmark of ependymomas is the perivascular pseudorosette. Well-formed true ependymal rosette.
- ✧ Ependymoma showing uniform ependymal tumour cells forming rosettes and canaliculi.
- ✧ On microscopic examination, ependymomas are composed of cells with regular, round to oval nuclei with abundant granular chromatin. Tumor cells may form round or elongated structures (rosettes, canals).
- ✧ Between the nuclei, there is a variably dense fibrillary background. Tumor cells may form gland-like round or elongated structures (rosettes, canals) that resemble the embryologic ependymal canal, with long, delicate processes extending into a lumen; more frequently present are perivascular pseudorosettes in which tumor cells are arranged around vessels with an intervening zone consisting of thin ependymal processes directed toward the wall of the vessel.
- ✧ The microscopic appearance of an ependymoma reveals a rosette pattern with the cells arranged about a central vascular space. It is histologically benign.

Medulloblastoma

- ✧ This tumor occurs predominantly in children and exclusively in the cerebellum.
- ✧ Neuronal and glial markers may be expressed, but the tumor is often largely undifferentiated.

Morphology.

- ✧ In children, medulloblastomas are located in the midline of the cerebellum, but lateral locations are more often found in adults. Rapid growth may occlude the flow of CSF, leading to hydrocephalus.
- ✧ The tumor is often well circumscribed, gray, and friable and may be seen extending to the surface of the cerebellar folia and involving the leptomeninges. On microscopic examination, medulloblastoma is usually extremely cellular, with sheets of anaplastic cells.

MENINGIOMAS

- ✓ Meningiomas are predominantly benign tumors of adults, usually attached to the dura, that arise from the meningotheelial cell of the arachnoid.
- ✓ Meningioma is the second most common primary intracranial neoplasm.
- ✓ Most cases are benign and slow growing, however certain subtypes show more aggressive behavior.
- ✓ This neoplasm most often occurs after 30 years of age. It occurs more frequently in women than in men.
- ✓ The neoplasm originates in arachnoidal cells of the meninges; the tumor is external to the brain and can often be successfully removed surgically.
- ✓ This neoplasm occurs most frequently in the convexities of the cerebral hemispheres and the parasagittal region; other common locations include the falx cerebri, sphenoid ridge, olfactory area, and suprasellar region.

This light tan colored **meningioma** involved the sphenoid ridge, another common location. The base of the sella is seen here, above which is the tan-colored pituitary gland. Above and on each side lateral to pituitary are the internal carotid and the anterior cerebral

arteries, above which slightly medially are the optic nerves



- ✓ Meningiomas may be found along any of the external surfaces of the brain as well as within the ventricular system, where they arise from the stromal arachnoid cells of the choroid plexus
- ✓ At medium power, this meningioma is composed of whorled nests of cells. A variety of patterns are possible
- ✓ Meningioma with a whorled pattern of cell growth and psammoma bodies

Morphology.

- ✓ Meningiomas are usually rounded masses with a well-defined dural base that compress underlying brain but are easily separated from it.
- ✓ Extension into the overlying bone may be present. The surface of the mass is usually encapsulated with thin, fibrous tissue and may have a bosselated or polypoid appearance.
- ✓ Meningioma. The tumor arises from the dura, is wellencapsulated, and is often cured by surgical removal.

METASTATIC TUMORS

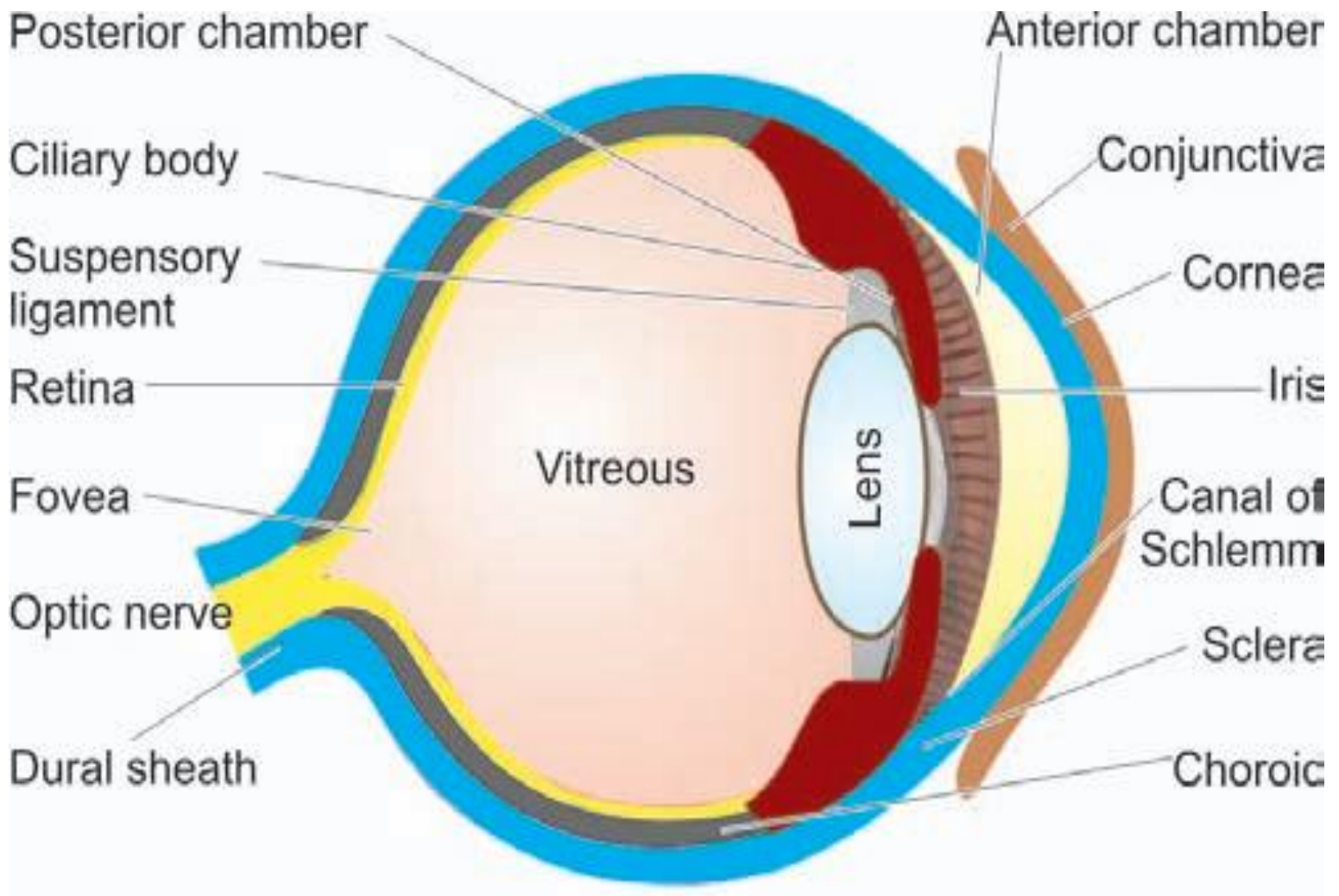
- ✓ Metastatic lesions, mostly carcinomas, account for approximately a quarter to half of intracranial tumors in hospital patients.
- ✓ The five most common primary sites are lung, breast, skin (melanoma), kidney, and gastrointestinal tract, accounting for about 80% of all metastases.
- ✓ Some rare tumors (e.g., choriocarcinoma) have a high likelihood of metastasizing to the brain, whereas other, more common tumors (e.g., prostatic carcinoma) almost never do even when they are metastatic to adjacent bone and dura.
- ✓ The meninges are also a frequent site of involvement by metastatic disease. Metastatic tumors present clinically as mass lesions and may occasionally be the first manifestation of the cancer.

Eye

Normal Structure

- The eyelids are covered externally by the skin and internally by conjunctiva which is reflected over the globe of the eye. The lacrimal glands which are compound racemose glands are situated at the outer upper angle of the orbit.
- The globe of the eye is composed of 3 layers: the cornea-sclera, choroid-iris, and retina.
- The cornea consists of stratified epithelium which may be regarded as continuation of the conjunctiva over the cornea. The subepithelial stroma consists of fibrous connective tissue, and the posterior endothelium-lined thin elastic membrane called Descemet's membrane.
- The sclera is composed of dense fibrous tissue which is thickest at the back of the eyeball.
- The choroid is the vascular membrane in contact with the sclera. The choroid becomes thickened anteriorly forming ciliary body, ciliary processes and contains ciliary muscle.
- The iris is the continuation of the choroid which extends in front of the lens. It is similar in structure to the choroid but contains pigment cells.
- The uveal tract consists of 3 parts—the choroid and ciliary body posteriorly, and the iris anteriorly. The retina is part of the central nervous system and corresponds in extent to the choroid which it lines internally.
- The retina is composed of a number of layers of cells and their synapses which are of 3 types—external photoreceptor cells (rods and cones), intermediate relay layer of bipolar cells, and internal layer of ganglion cells with their axons running into the central nervous system.
- The lens is the biconvex mass of laminated transparent tissue with elastic capsule.

- The anterior chamber is the space filled with the aqueous humour, and is bounded by the cornea in front and the iris behind, with anterior surface of the lens exposed in the pupil.
- The posterior chamber containing aqueous humour is the triangular space between the back of the iris, the anterior surface of the lens and ciliary body forming its apex at the pupillary margin.
- The vitreous chamber is the large space behind the lens containing gelatinous material, the vitreous humour.
- The main function of the eye is visual acuity which depends upon a transparent focussing system comprised by the cornea, lens, transparent media consisting of aqueous and vitreous humours, and a normal retinal and neural conduction system.



The Eyelids

The important conditions affecting the eyelids include:

Blepharitis

inflammation of the eyelids. It is common and sometimes produces an acute, red, tender, inflammatory mass.

Hordeolum (or sty)

refers to an acute, inflammatory, focal lesion of the eyelid. Acute inflammation involving the Meibomian glands is termed an internal hordeolum, whereas acute folliculitis of the glands of Zeis is an external hordeolum.

Chalazion

a granulomatous inflammation centered around the meibomian glands or the glands of Zeis. It is thought to represent a reaction to extruded lipid secretions and usually produces a painless swelling in the eyelid.

Xanthelasma

a yellow plaque of lipid-containing macrophages, usually involving the nasal aspect of the eyelids. It is often seen in older persons and patients with disorders of lipid metabolism (e.g., familial hypercholesterolemia, primary biliary cirrhosis).

The Orbit

Exophthalmos or Proptosis

- Is Abnormal Forward Protrusion of the Eyeball. The term exophthalmos is used mainly when the condition is bilateral; proptosis refers to a unilateral protrusion of the eye. Numerous conditions cause forward protrusion of the
- eye. The most common cause is thyroid disease, followed by orbital dermoid cysts and hemangiomas. Other orbital conditions can cause proptosis: various inflammatory lesions, lymphomas, developmental anomalies, vascular problems and neoplasms. Proptosis also results from lesions of the paranasal sinuses and intracranial cavity.

Exophthalmos of Hyperthyroidism

- Continues Despite Treatment
- Exophthalmos caused by Graves disease may precede or follow other manifestations of thyroid dysfunction. Exophthalmos resulting from thyroid disease usually occurs in early adult life, especially in women (female-to-male ratio, 4:1). It may be severe and progressive, particularly in middle life, when exophthalmos no longer correlates well with the state of thyroid function.
- Dysthyroid exophthalmos may be associated with edema of the eyelids, chemosis (conjunctival edema) and limitation of ocular motion.

The Conjunctiva

Conjunctival Hemorrhage

May Follow Blunt Trauma, Anoxia or Severe Coughing Conjunctival hemorrhages also occur spontaneously, often first noted on arising after sleep. They do not extend into the cornea because of the barrier imposed by the close apposition of corneal epithelium to the underlying substantia propria.

Conjunctivitis

Chlamydia is responsible for a purulent conjunctivitis (inclusion blennorrhea) that develops in newborns, who become infected during passage through the birth canal. The infection is also acquired by swimming in nonchlorinated pools (swimming pool conjunctivitis) or from contact with discharges of infected urethra or cervix.

In adults and older children, Chlamydia causes a chronic follicular conjunctivitis with focal lymphoid hyperplasia (inclusion conjunctivitis). In contrast to trachoma, the lower tarsal conjunctiva is involved. Scarring and necrosis do not develop, and keratitis is rare and mild.

Ophthalmia neonatorum

Ophthalmia neonatorum is a severe, acute conjunctivitis with a copious purulent discharge, especially in the newborn, caused by *Neisseria gonorrhoeae*. The infection, which is a common cause of blindness in some parts of the world, is complicated by corneal ulceration, perforation and scarring and panophthalmitis. Infants usually become infected while passing through the birth canal of an infected mother. Other causative organisms for ophthalmia neonatorum include other pyogenic bacteria and *C. trachomatis*. Today, newborns are usually routinely treated with 5% Betadine eye drops.

CONJUNCTIVITIS AND KERATOCONJUNCTIVITIS.

Conjunctiva and cornea are constantly exposed to various types of physical, chemical, microbial (bacteria, fungi, viruses) and allergic agents and hence prone to develop acute, subacute and chronic inflammations.

In the acute stage, there is corneal oedema and infiltration by inflammatory cells, affecting the transparency of the cornea. In the more chronic form of inflammation, there is proliferation of small blood vessels in the normally avascular cornea and infiltration by lymphocytes and plasma cells (pannus formation).

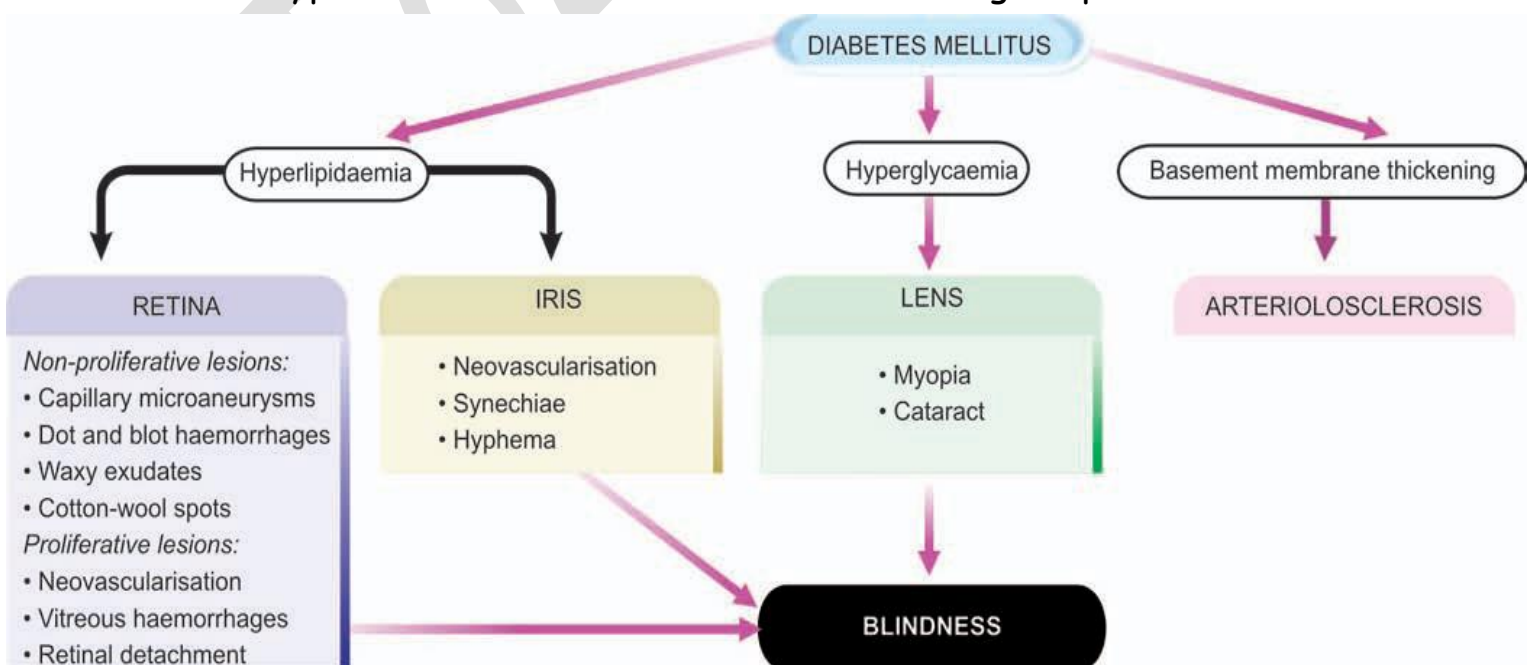
INFLAMMATORY CONDITIONS

- Inflammatory conditions of the eye are designated according to the tissue affected. 'Uveitis' is the commonly used term for the ocular inflammation of the uveal tract which the most vascular tissue of the eye. However, specific designation is used for the type of tissue of eye inflamed.

VASCULAR LESIONS

DIABETIC RETINOPATHY.

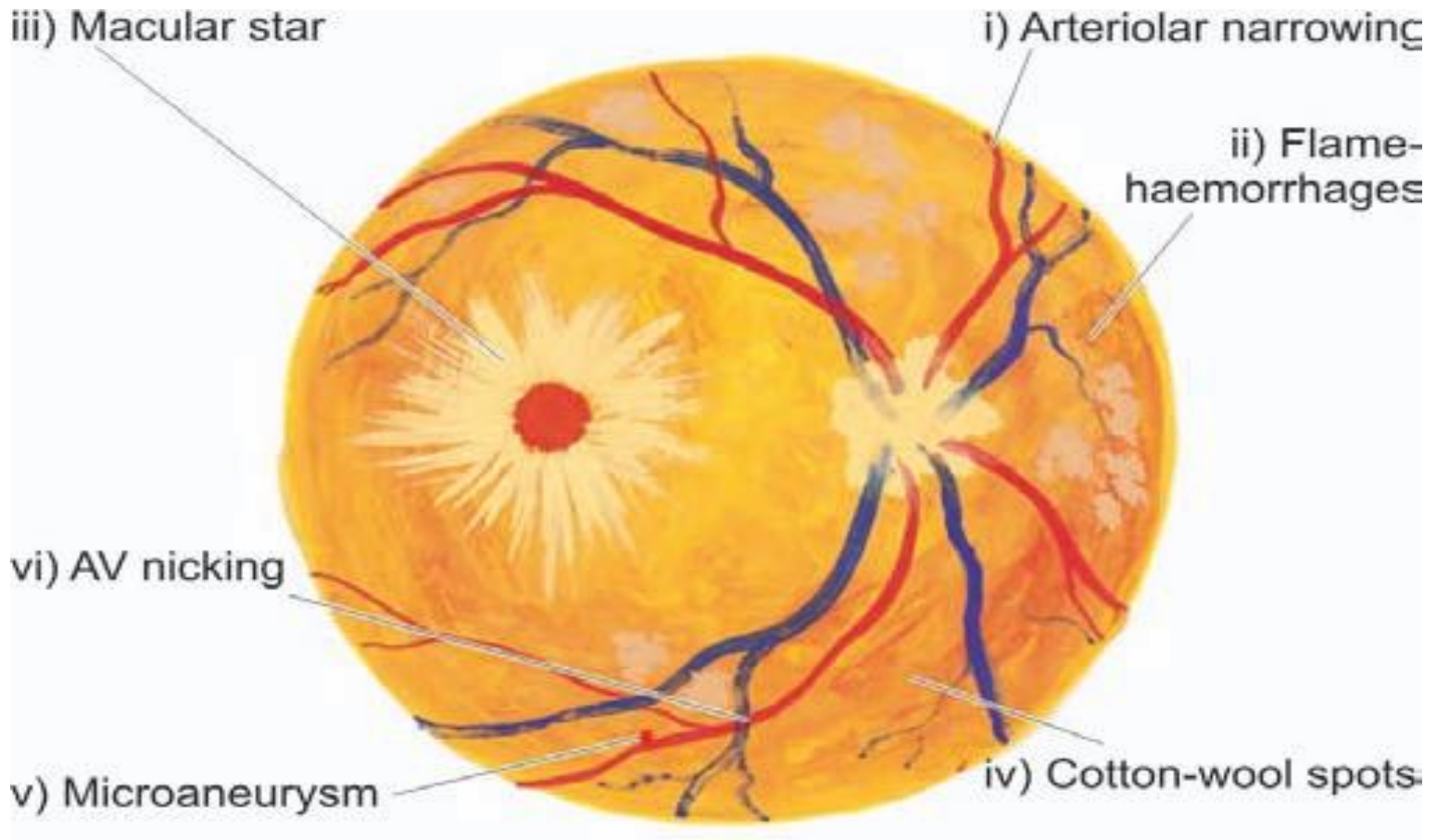
- Diabetic retinopathy is an important cause of blindness. It is related to the degree and duration of glycaemic control. The condition develops in more than 60% of diabetics 15-20 years after the onset of disease, and in about 2% of diabetics causes blindness.
- Other ocular complications of diabetes include glaucoma, cataract and corneal disease. Most cases of diabetic retinopathy occur over the age of 50 years.
- The risk is greater in type 1 diabetes mellitus than in type 2 diabetes mellitus, although in clinical practice there are more patients of diabetic retinopathy due to type 2 diabetes mellitus because of its higher prevalence. Women are



more prone to diabetes as well as diabetic retinopathy. Diabetic retinopathy is directly correlated with Kimmelstiel-Wilson nephropathy.

HYPERTENSIVE RETINOPATHY

- In hypertensive retinopathy, the retinal arterioles are reduced in their diameter leading to retinal ischaemia. In acute severe hypertension as happens at the onset of malignant hypertension and in toxæmia of pregnancy, the vascular changes are in the form of spasms, while in chronic hypertension the changes are diffuse in the form of onion-skin thickening of the arteriolar walls with narrowing of the lumina.
- Features of hypertensive retinopathy include the following:
 - 1) Variable degree of arteriolar narrowing due to arteriolosclerosis.
 - 2) 'Flame-shaped' haemorrhages in the retinal nerve fibre layer.
 - 3) Macular star i.e. exudates radiating from the centre of macula.
 - 4) Cotton-wool spots i.e. fluffy white bodies in the superficial layer of retina.
 - 5) Microaneurysms.
 - 6) Arteriovenous nicking i.e. kinking of veins at sites where sclerotic arterioles cross veins.
 - 7) Hard exudates due to leakage of lipid and fluid into macula.



Other Lesions

RETINAL DETACHMENT.

- Retinal detachment is the separation of the neurosensory retina from the retinal pigment epithelium. It may occur spontaneously in older individuals past 50 years of age, or may be secondary to trauma in the region of head and neck. Normally, the rods and cones of the photoreceptor layer are interdigitated with projections of the retinal pigment epithelium, but the two can separate readily in some disease processes.

CATARACT.

- The cataract is the opacification of the normally crystalline lens which leads to gradual painless blurring of vision. The various causes of cataract are: senility, congenital (e.g. Down syndrome, rubella, galactosaemia), traumatic (e.g. penetrating injury, electrical injury), metabolic (e.g. diabetes,

hypoparathyroidism), and associated with drugs (e.g. longterm corticosteroid therapy), smoking and heavy alcohol consumption.

Retinoblastoma

- This is the most common malignant ocular tumour in children. It may be present at birth or recognised in early childhood before the age of 4 years.
- Retinoblastoma has some peculiar features. About 60% cases of retinoblastoma are sporadic and the remaining 40% are familial. Familial tumours are often multiple and multifocal and transmitted as an autosomal dominant trait by retinoblastoma susceptibility gene (RB) located on chromosome 13.
- Such individuals have a higher incidence of bilateral tumours and have increased risk of developing second primary tumour, particularly osteogenic sarcoma. Retinoblastoma may occur as a congenital tumour too.

MORPHOLOGIC FEATURES.

- **Grossly**, the tumour characteristically appears as a white mass within the retina which may be partly solid and partly necrotic. The tumour may be endophytic when it protrudes into the vitreous, or exophytic when it grows between the retina and the pigment epithelium.
- **Microscopically**, the tumour is composed of undifferentiated retinal cells with tendency towards formation of photo-receptor elements. In the better differentiated area, the tumour cells are characteristically arranged in rosettes.
- The rosettes may be of 2 types—**Flexner-Wintersteiner** rosettes characterised by small tumour cells arranged around a lumen with their nuclei away from the lumen, and **Homer-Wright** rosettes having radial arrangement of tumour cells around the central neurofibrillar structure .
- The tumour shows wide areas of necrosis and calcification and dissemination in all directions—into the vitreous, under the retina, into the optic nerve and even into the brain.

SJÖGREN'S SYNDROME.

Sjögren's syndrome is characterised by triad of keratoconjunctivitis sicca, xerostomia (sicca syndrome) and rheumatoid arthritis. The condition occurs due to immunologically-mediated destruction of the lacrimal and salivary glands along- with another autoimmune disease

MIKULICZ'S SYNDROME.

This is characterised by inflammatory enlargement of lacrimal and salivary glands. The condition may occur with Sjögren's syndrome, or with some diseases like sarcoidosis, leukaemia, lymphoma and macroglobulinaemia.

Revision Questions

Section A (Scenario Questions)

1. A baby is stillborn at 34 week' gestation to a 33-year-old gravida 3 para 2 woman, whose two previous pregnancies had resulted in normal term infants. The baby is observed to be hydropic. Autopsy on the fetus reveals marked organomegaly, and the brain has extensive necrosis in a periventricular pattern. Which of the following congenital infections is most likely to have accounted for these findings?

- a. *Listeria monocytogenes*
 - b. Human immunodeficiency virus
 - c. Herpes simplex virus
 - d. Cytomegalovirus
2. The gross appearance shown was observed on sectioning the brain from a 45-year-old male who died from acute meningitis with *Streptococcus pneumoniae*. Based on this appearance, which of the following complications has likely resulted from his infection?
- a. Transtentorial herniation
 - b. Hydrocephalus
 - c. Abscess formation
 - d. Subarachnoid hemorrhage
 - e. Laminar cortical necrosis
3. Postmortem examination on the body of 60-year-old man showed an infiltrating neoplasm invading the cerebral hemispheres and crossing the midline, with areas of necrosis. Histology demonstrates hemorrhage and a pseudopalisade arrangement of tumor cells. Abnormal blood vessels are also present. Before patient died there had been a recent onset of headache, seizures, and mental status changes. The tumor is most likely a(n)
- a. ependymoma.
 - b. glioblastoma multiforme.
 - c. meningioma.
 - d. oligodendroglioma.
4. Following a bar fight, a 22-year-old man is brought unconscious to the emergency department. Several minutes earlier, he had been hit on the head with a heavy iron club and had been briefly unconscious, but had then apparently recovered. One or two minutes later, he had again lost consciousness. Which of the following is the most likely diagnosis?
- a. Epidural hematoma
 - b. Subarachnoid hemorrhage
 - c. Subdural hematoma

d. Transient ischemic attack

5. A 2-year-old child presents with fever, headache, prostration, and nuchal rigidity. The CSF is cloudy, and microscopic examination reveals innumerable neutrophils. The CSF protein is increased, and glucose is decreased. The most likely etiologic agent is
- Escherichia coli.
 - Haemophilus influenzae.
 - group B streptococci.
 - Streptococcus pneumoniae.
 - Staphylococcus aureus.
6. The figures shown here are representative of the gross and microscopic appearance of an autopsy specimen from a 55-year-old woman who had a 1-year history of progressive headache and seizures leading to aphasia. The diagnosis is
- ependymoma.
 - glioblastoma multiforme.
 - medulloblastoma.
 - meningioma.
 - oligodendroglioma.
7. Which ONE of the following is not a common site of hypertensive intracerebral haemorrhage?
- Basal ganglia.
 - Cerebellum.
 - Internal capsule.
 - Occipital poles.
 - Pons.
8. Which ONE of the following is not a common site of a berry aneurysm?
- Basilar artery.
 - Bifurcation of the middle cerebral artery.
 - Junction of anterior communicating artery and anterior cerebral artery.

- d. Junction of internal carotid artery and posterior communicating artery.
 - e. Vertebral artery.
9. If the following events were placed in chronological order which would come fourth?
- a. Abscess in temporal lobe.
 - b. Chronic otitis media.
 - c. Inadequate antibiotic therapy.
 - d. Invasion of venous channels by bacteria.
 - e. Osteitis of the tegmen tympani.
10. Which ONE of the following is the commonest intracerebral neoplasm?
- A. Astrocytoma.
 - B. Ependymoma.
 - C. Meningioma.
 - D. Oligodendroglioma.
 - E. Secondary carcinoma.

Section B (True or False)

1. Analyse the following statements and decide whether they are true or false:-
- a. A subdural hematoma is hemorrhage resulting from laceration of the bridging veins.
 - b. Choroid plexus is the central nervous system (CNS) organ responsible for producing cerebrospinal fluid.
 - c. A subdural hematoma is venous hemorrhage between the skull and dura.
 - d. inflammation of the spinal cord is named myelitis.
 - e. Most infectious agents reach the brain and the meninges through the lymphatics.
2. Analyse the following statements and decide whether they are true or false:-
- a. Hepatic Hemangiomas can cause abdominal symptoms and even hemorrhage into the peritoneal cavity.

- b. Acinar cell injury and duct obstruction are the major causes of acute pancreatitis.
- c. Carcinoma of the pancreas is usually adenocarcinoma carcinoma.
- d. Microscopically, hepatic hemangiomas similar to cavernous hemangiomas are found elsewhere.
- e. Hepatic Hemangiomas are rare tumors of the Liver.

3. Analyse the following statements and decide whether they are true or false-:

- a. Hypertension is the most common underlying cause of primary brain parenchymal hemorrhage
- b. Subarachnoid haemorrhage is commonly atherosclerotic aneurysms in origin
- c. Atherosclerotic aneurysm causes cerebellar hemorrhage
- d. Choroid plexus is the central nervous system (CNS) organ responsible for producing cerebrospinal fluid.
- e. The frontal lobe is responsible for voluntary movement, intellect, personality and memory..

4. Analyse the following statements and decide whether they are true or false-:

- a. Spontaneous (nontraumatic) intraparenchymal hemorrhages occur most commonly in middle to late adult life
- b. vascular malformation is the most common underlying cause of primary brain parenchymal hemorrhage
- c. Haemorrhage into the brain may be traumatic, non-traumatic, or spontaneous.
- d. Subarachnoid haemorrhage is commonly traumatic in origin.
- e. Hypertension is one of the causes of spontaneous intracranial haemorrhage

5. Analyse the following statements and decide whether they are true or false-:

- a. An aneurysm is a circumscribed, segmental dilatation of an artery.
- b. Berry aneurysms are small saccular aneurysms; typically found in and around the circle of Willis at the base of the brain.
- c. The rupture of Atherosclerotic aneurysms lead to intraventricular hemorrhage.

- d. Berry aneurysms are most often asymptomatic and rarely rupture.
- e. Long-standing hypertension cause microscopic Charcot-Bouchard aneurysms.

6. Analyse the following statements and decide whether they are true or false-:

- a. The white matter beneath the cerebral cortex is composed of neurons.
- b. A contusion describes damage to the cortex in the form of a bruise or laceration, which is not associated with the hemorrhage.
- c. Contusions is life-threatening if complicated by edema and hemorrhage.
- d. Meningococcemia associated with purpuric skin lesions and is sometimes complicated by the Waterhouse-Friderichsen syndrome.
- e. Leptomeningitis is the commonest form of bacterial infection in the CNS

7. Viral infections (true or false)

- a. Infections spread to the CNS by the haematogenous route or by retrograde neural transport
- b. Viral meningitis is a rare, self-limiting illness with characteristic CSF changes
- c. Encephalitis is less frequent, but may result in death or severe disability
- d. Reactivation of a latent viral infection (e.g. herpes zoster) may damage the CNS
- e. CNS involvement in HIV infection is common and often accompanied by other viral, bacterial or parasitic infections

8. Analyse the following statements and decide whether they are true or false-:

- a. Viral meningitis is aseptic meningitis.
- b. Viral meningitis may occur as part of a systemic viral disease (e.g., varicella, mumps, and measles),
- c. Viral meningitis may be limited to the CNS.
- d. Viral meningitis is rare associated with encephalitis.

9. Analyse the following statements about Gilbert Syndrome and indicate whether they are true or false:

- a. Gilbert syndrome develops into liver cirrhosis.
- b. It is typically diagnosed during the investigation of jaundice that has appeared in the course of some common disease.
- c. Syndrome invariably develop bilirubin encephalopathy.
- d. Jaundice may also be precipitated by minor stress.
- e. Gilbert syndrome is treated by liver transplantation.

10. Analyse the following statements and decide whether they are true or false-:

- a. Disseminated intravascular coagulation is the systemic (distant) complications of acute hemorrhagic pancreatitis.
- b. Melanomas develop often in congenital melanocytic nevi.
- c. Carcinoma of the bile duct: Increased incidence in Crohn's disease.
- d. Adenocarcinoma is the most common tumor of the gallbladder.
- e. Hepatic Hemangiomas are ordinarily small and asymptomatic.

11. Analyse the following statements on meningitis and decide whether they are true or false-:

- a. Infection may be limited to the subarachnoid space.
- b. Infection may spread into the brain.
- c. Infection may be classified as acute or chronic.
- d. Infection may be aseptic.
- e. The appearance of the cerebrospinal fluid of viral meningitis is purulent.

12. Analyse the following statements on cerebral abscesses and decide whether they are true or false-:

- a. General signs and symptoms of infection and increased intracranial pressure.
- b. Abscesses may cause hemiplegia, seizures, and coma.
- c. CSF finding is specific and diagnostic in the majority of cases.
- d. Computed tomography (CT) shows typical ring-enhancing lesions.
- e. Without surgery and chemotherapy, it is not fatal in most cases.

13. Analyse the following statements on brain neoplasms and decide whether they are true or false-:

- a. Most brain tumors are malignant.
- b. In general, brain tumors do not metastasize
- c. Most brain tumors originate from glia cells.
- d. Meningiomas are malignant tumors. F
- e. Schwannomas are benign intracranial tumors.

14. Analyse the following statements and decide whether they are true or false-:

- a. Atherosclerotic aneurysms are situated primarily in large cerebral arteries
- b. Mycotic aneurysms result from septic emboli that originate in infected cardiac valves and can cause arterial rupture, cerebral abscess, or meningitis.
- c. Charcot-Bouchard aneurysms: formation and rupture of these aneurysms occurs most commonly along the trunk of the vessel, rather than at points of bifurcation.
- d. Spontaneous cerebral hemorrhage generally occurs as a consequence of longstanding hypertension.
- e. A coup injury occurs at the site of impact, whereas a contrecoup injury is contralateral to the site of initial injury.

15. Analyse the following statements and decide whether they are true or false-:

- a. Contusions may be life-threatening if complicated by edema and hemorrhage, which predispose to transtentorial herniation.
- b. A concussion is a transient loss of consciousness due to trauma that causes a rapid torque on the brainstem, leading to a paralysis of neurons of the reticular formation.
- c. Epidural Hematoma Results from Bleeding Between the Skull and Dura
- d. Subarachnoid hemorrhage may occur following trauma or rupture of a berry aneurysm in the Circle of Willis
- e. Vasculitis and tumors may cause subarachnoid hemorrhage.

16. Analyse the following statements on tuberculosis and decide whether they are true or false-:
- a. Choroid plexus is the central nervous system (CNS) organ responsible for producing neurons.
 - b. A subdural hematoma is venous hemorrhage between the skull and dura.
 - c. A subdural hematoma is hemorrhage resulting from laceration of the bridging veins.
 - d. Pyogenic meningitis in older infants and young children is most frequently caused by *Listeria*
 - e. Most infectious agents reach the brain and the meninges through the lymphatics.
17. Analyse the following statements and indicate whether they are true or false:
- a. Leptomeningitis is the rare form of bacterial infection in the CNS.
 - b. Chronic meningitis may be a feature of tuberculosis.
 - c. Cryptococcal Meningitis Occurs Most Frequently in Immunocompromised Persons
 - d. An epidural hematoma is an arterial hemorrhage between the dura and the skull
 - e. Hydrocephalus is one of complications of meningitis.
18. Analyse the following statements and indicate whether they are true or false:
- a. Viral Meningitis is usually benign and resolves without sequelae.
 - b. Herpes Simplex Virus is the Most Common Cause of Nonepidemic Encephalitis
 - c. Duret hemorrhages are hemorrhages within the cerebellum.
 - d. In adults, the majority of intracranial tumors are infratentorial.
 - e. Meningioma is the second most common primary intracranial neoplasm.

Section C (Evaluation Questions)

1. What are the most important forms of intracranial bleeding?
2. What is epidural hematoma ?
3. What is subdural hematoma?
4. List the most common aneurysms found in the CNS.
5. What are cerebral arteriovenous malformations?

6. What are the pathologic consequences of global cerebral ischemia?
7. What are the most important causes of occlusion of cerebral arteries?
8. What are the most important infectious diseases of the CNS ?
9. List the most common causes of acute bacterial meningitis.
10. Describe the gross and microscopic findings in acute bacterial meningitis.
11. List the most important causes of chronic meningitis.
12. Describe the pathologic findings in chronic meningitis.
13. List the most important causes of viral meningitis.
14. What are the typical clinical features of acute and chronic meningitis?
15. List the complications of meningitis.
16. List key facts about the pathogenesis and pathology of cerebral abscesses.
17. List key facts about the clinical presentation of cerebral abscess.
18. List key facts about viral encephalitis.
19. List key facts about poliomyelitis.
20. How does HIV-1 infection affect the CNS?
21. What are the typical pathologic findings in HIV encephalopathy?
22. How common are brain tumors?
23. List the most common primary brain tumors.
24. How are tumors of the CNS classified topographically?
25. List the pathogenesis of clinical signs and symptoms caused by brain tumors.
26. Do brain tumors metastasize?
27. Can benign brain tumors cause death?
28. List key facts about ependymoma.
29. What is medulloblastoma?
30. List key facts about meningioma.
31. Which tumors tend to metastasize to the brain?

Section D (Matching)

a	<i>Mycotic aneurysms</i>	A	<i>Most often, they are asymptomatic and rarely rupture.</i>
b	<i>Berry aneurysms</i>	B	<i>Long-standing hypertension</i>
c	<i>Arteriovenous malformation.</i>	C	<i>result from septic emboli that originate in infected cardiac valves</i>
		D	<i>Located around the circle of Willis</i>
		E	<i>A disorganized collection of arteries and veins is seen within the substance of the brain.</i>

a	<i>Acute subdural haematoma.</i>	A	<i>Associated with cerebral contusions and lacerations in many cases.</i>
b	<i>Chronic subdural haematoma.</i>	B	<i>Due to rupture of an aneurysm.</i>
c	<i>Extradural haematoma.</i>	C	<i>May present weeks after trivial injury.</i>
		D	<i>Scarring of frontal poles.</i>

		<i>E</i>	<i>Usually due to a tear of the middle meningeal artery.</i>
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